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Managing Perishable Inventory Systems with Multiple Demand Classes

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Abstract

In this paper, we study a multi-period stochastic perishable inventory system with multiple demand classes that require products of different ages. The firm orders the product with a positive leadtime and sells it to multiple demand classes, each only accepting products with remaining lifetime longer than a threshold. In each period, after demand realization, the firm decides how to allocate the on-hand inventory to different demand classes with different backorder or lost-sale cost. At the end of each period, the firm can dispose inventory of any age. We formulate this problem as a Markov decision process and characterize the optimal ordering, allocation, and disposal policies. When unfulfilled demand is backlogged, we show that the optimal order quantity is decreasing in the inventory levels and is more sensitive to the inventory level of fresher products, the optimal allocation policy is a sequential rationing policy, and the optimal disposal policy is characterized by a set of thresholds. For the lost-sale case, we show that the optimal allocation and disposal policies have the same structure but the optimal ordering policy may be different. Based on the structure of the optimal policy, we develop an efficient heuristic that is at most 4% away from the optimal cost in our numerical examples. Using numerical studies, we show that the ordering and allocation policies are close to optimal even if the firm cannot intentionally dispose products. Moreover, ignoring the difference between demand classes and using a simple allocation policy (e.g., FIFO) can significantly increase the total cost. We examine how the firm can improve the control of perishable items and show that the benefit of decreasing the leadtime is more significant than that of increasing the lifetime of the products or that of decreasing the acceptance threshold of the demand. The analysis is extended to systems with age dependent disposal cost and stochastic supply.

Keywords: Perishable inventory systems; multiple demand classes; stochastic optimal control; demand fulfillment and rationing; multimodularity

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1 Introduction

The control of perishable products is becoming increasingly important in supply chain management. Mismanagement of perishable products can lead to significant loss of profit for companies in the grocery retailing industry and significant spoilage of drugs in hospitals. For example, according to Buzby et al. (2014), the estimated total value of food loss at the retail level in the United States was $53.8 billion in 2010. The 2011 National Blood Collection and Utilization Survey reported that 4.3% of all components of blood processed for transfusion were outdated in 2011 in the USA. Recognizing the economic and social importance of perishable products, researchers have conducted substantial work on perishable inventory problems. However, the perishable inventory problem with multiple demand classes with different requirements for the age of products, which is the focus of this paper, has attracted only limited attention.

Suppliers of perishable products usually face several classes of customers who have their own requirements. For example, consider a processed meat producer facing several customer types including a retailer and a fast food restaurant. At the retailer store, the meat products may remain on the shelf for several days before being sold and customers want to ensure that they can use the products for a period of time. Thus, the retailer expects to receive fresh products to ensure there is enough time till the units are expired. However, the restaurant may not be as sensitive to the age of products as the retailer since the products are used to make food for direct consumption, and as a results, it may accept products with a shorter remaining lifetime. Given the different customers' requirements, the supplier may provide price incentives to customers and charge different prices depending on the remaining shelf life to ensure the older units are sold before being expired. In addition, at the end of each day, the supplier may donate the surplus of its stock to a local food bank, where products will be consumed fast. Another example is a blood product supply chain where hospitals need fresh blood products for specific patients or operations. For example, patients with a major bleeding caused by trauma or surgery can be transfused with platelets of any age, while patients with a platelet function disorder, such as leukemia, need to be transfused with fresh platelets that are stocked for at most 3 days (see, e.g., Haijema et al. 2005 and Deniz et al. 2010). Similarly, suppliers of many other perishable items, such as dairy, flowers, and pharmaceutical drugs, may deliver products of different ages to retailers in different geographical

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locations. Retailers located far from the supplier usually require products with longer remaining lifetime since the transit time is longer. Local retailers may accept products that are relatively older.

Motivated by the above examples, in this paper, we examine how ordering and allocation of products with different ages to different demand classes affect the total cost of the system. Specifically, we consider the joint ordering, allocation, and disposal policy for a multi-period stochastic perishable inventory system. The firm orders the product from a supplier that has ample capacity and delivers the ordered quantities after a fixed leadtime, so that the firm receives products with remaining lifetime n. The firm sells the product to multiple demand classes, each only accepting products with remaining lifetime longer than a threshold. In each period, after demand realization, the firm decides how to allocate the on-hand inventory to the different demand classes. Unfulfilled demand is backordered and different demand classes may have different backorder costs. At the end of each period, the firm can dispose inventory of any age. The total cost we consider includes holding, shortage, and disposal over the entire planning horizon. We formulate this problem as a Markov decision process and show that the optimal cost function is multimodular. This allows us to derive the optimal ordering, allocation, and disposal policies.

In this paper, we make the following key contributions:

(1) As far as we know, our paper is the first to examine the optimal joint ordering, allocation, and disposal decision for a perishable inventory system with multiple demand classes that require products of different ages. Previous literature that studies perishable inventory systems with multiple demand classes usually focuses on identifying effective heuristics for a certain type of policy. For example, Haijema et al. (2005, 2007, 2009) use simulation to construct several simple heuristics for the ordering policy that are nearly optimal under a given allocation and disposal policies. Civelek et al. (2015) propose a critical level (or protection level) allocation policy under some practical replenishment policies and show that this policy leads to superior performance compared to existing heuristics. However, none of them examine the optimal joint ordering, allocation, and disposal policy or propose heuristics for the joint policy. Our paper fills in this gap in the literature.

We characterize the structure of the optimal ordering, allocation, and disposal policies for the case with general $m(\leq n)$ demand classes. In particular, we show that, as expected, the optimal order quantity is decreasing in the inventory levels and is more sensitive to the inventory of the

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fresher products. We establish that the optimal allocation policy is a sequential rationing allocation policy: first satisfy the demand with the highest priority, then the demand with the second highest priority, and so on, such that while fulfilling each demand class, it is optimal to ration fresher products to certain thresholds for future demand with higher priority, and use the rest to fulfill demand of that class as much as possible following the First-In First-Out (FIFO) policy. We find that the optimal disposal policy is characterized by $n-1$ thresholds, where each threshold specifies the reservation level for products with a particular remaining lifetime. At the end of each period, only if the inventory levels are larger than the thresholds, it is optimal to dispose the surplus.

We note that Chen et al. (2014) study the joint pricing and inventory control problem for perishable products with a single demand class. In that paper the focus is on characterization of the joint inventory and pricing decisions. Subject to the assumption that each demand class accepts products of all ages, they claim that the results of the single demand class model can be extended to models with multiple demand classes. In contrast, in this paper, we assume that different demand classes require products of different ages. Consequently, the optimal allocation policies are different. For example, Chen et al. (2014) indicate that "for any given total amount allocation, it is always optimal to meet the demands to the maximum extent from the highest priority to the lowest priority"[1] and thus the optimal allocation policy can be characterized by this single amount. This is not true in our model. Instead, we show the optimal allocation policy is a sequential rationing policy. Moreover, as we discuss below, the approach in Chen et al. (2014) cannot be applied in our model.

(2) We show that the structure of the optimal policy is robust with respect to different assumptions and the optimal ordering and allocation policies are close to optimal even if the firm cannot intentionally dispose products. Moreover, we show that it is important to consider the difference in the requirements on the age of acceptable products between demand classes. Ignoring this fact and using simple policies (e.g., FIFO) can significantly increase the total cost. This also suggests that the heuristics proposed in Haijema et al. (2005, 2007, 2009) may perform poorly when compared to the optimal joint policy. Based on the structure of the optimal joint policy, we develop an effective and computationally efficient heuristic which is at most 4% away from the optimal cost in our numerical examples.

[1]See page ec6 of the online supplement of Chen et al. (2014).

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(3) We examine how the firm can improve the control of perishable items. Clearly, the firm can reduce its total cost and the number of products disposed by decreasing the leadtime, increasing the lifetime of the products, or increasing customers' willingness to accept older products. We show that, among the above three strategies, decreasing leadtime is the most efficient (with a potential cost benefit of 20% in our numerical examples). Increasing the lifetime is more efficient than increasing customers' willingness to accept older products.

(4) We also make a contribution from the analysis perspective. Characterizing the structure of the optimal policy and deriving the comparative statics of the optimal thresholds and the optimal order quantity rely on the property of "multimodularity". This concept (or an equivalent concept "$L^\natural$-convexity") has been used to characterize the structure of the optimal policy for many inventory systems; see, for example, Zipkin (2008), Huh and Janakiraman (2010), Chen et al. (2014), and Li and Yu (2014). In these papers, to show that the optimal value function is multimodular or $L^\natural$-convex, the idea is to first show that multimodularity or $L^\natural$-convexity is preserved for constraints with a certain structure and then change the constraints in the original dynamic program to new ones that have this structure. However, when there are multiple demand classes that require products of different ages, this approach does not work. Therefore, we use the definition of multimodularity to show that after state transformation, the new function is submodular. This requires showing that submodularity is preserved under minimization with a non-lattice constraint, which is not trivial.

2 Literature Review

There is a large body of literature on inventory problems with perishable products. Broadly speaking, there are two streams in this literature, periodic review and continuous review; see the recent reviews by Nahmias (1982, 2011) focusing on periodic review, Baron (2010) focusing on continuous review, and Karaesmen et al. (2011) that includes both. Our paper belongs to the periodic review stream and we next focus on it.

In a setting with zero leadtime and lifetime of two periods, Nahmias and Pierskalla (1973) first study the optimal ordering policy for a perishable inventory system with a single stochastic demand class. Then, Nahmias (1975) and Fries (1975) characterize the structure of the optimal policy for the

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general lifetime problem with zero leadtime, in backorder models and lost-sales models, respectively. They show that the optimal ordering quantity depends on the levels of on-hand inventory of different ages. Because of the complexity and the state dependency, implementation of the optimal policy is difficult. Therefore, several simple and effective heuristic policies were proposed and examined, for example, NIS policy (Brodheim et al. 1975), TIS policy (Nahmias 1976, 1977; Nandakumar and Morton 1993; Cooper 2001; Haijema 2011), proportional-balancing policy (Chao et al. 2015), dual-balancing policy (Chao et al. 2015; Zhang et al. 2016), and truncated-balancing policy (Zhang et al. 2016), among others. All these papers make two common assumptions: there is a single class of demand and only the excess inventory that expires at the end of each period is disposed.

Recently, there is a growing body of literature on perishable inventory systems that allows inventory of any age to be salvaged. Xue et al. (2012) study a perishable inventory model where the surplus of stock can be sold (i.e., disposed) in a secondary market with some salvage value. They characterize the structure of the optimal policy and then propose a heuristic. By employing the concept of multimodularity, Li and Yu (2014) revisit the same problem and show that the optimal policy is monotone with bounded sensitivity. Chen et al. (2014) consider the joint pricing and inventory control problem for this model and allow for positive lead time. Haijema (2014) numerically investigates cost reductions that can be achieved by an optimal stock-age dependent ordering, issuing, or disposal policy. Li et al. (2016) study the problem of joint replenishment and clearance sales of perishable goods under a Last-In First-Out (LIFO) issuing rule and show that the optimal policy for clearance sales is characterized by two thresholds for each age group of inventory. All the papers discussed focus on a single demand class.

There is limited literature that studies perishable inventory systems with multiple demand classes. Much of the literature either considers a single period problem (see, for example, Ishii 1993; Ishii and Nose 1996) or focuses on identifying effective heuristics for a certain type of policy (see, for example, Haijema et al. 2005, 2009; Deniz et al. 2010; Civelek et al. 2015). None of them examine the optimal joint ordering, allocation, and disposal policy or propose heuristics for the joint policy. Our paper fills in this gap in the literature.

Finally, our work is related to the literature on inventory rationing and capacity management with multiple capacity types and multiple demand classes. Topkis (1968) considers the allocation of non-perishable inventory to multiple demand classes in a multi-period setting and shows that

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a static rationing policy is optimal. Shumsky and Zhang (2009) consider a multi-period capacity allocation model with single-step upgrading. They show that it is optimal to use any available capacity to satisfy same-class demand, and then upgrade customers until capacity reaches a protection limit. Yu et al. (2015) extends this model to allow general upgrading and show that a sequential upgrading policy is optimal. A common assumption in this literature is that there is no replenishment opportunity during the selling season. However, our model allows the firm to replenish inventory in each period.

3 Model with Two Demand Classes

We consider a single item multi-period stochastic inventory control problem where a firm serves two demand classes. The supplier has ample capacity and delivers the order quantities after a leadtime L so that orders placed in period t arrive in period $t+L$. The planning horizon consists of $T+L \geq 1$ discrete time periods. Time $t = 1$ is the first period and time $t = T + L$ is the last period. The product is perishable and has a lifetime of $n + L$ periods. Thus, products have remaining lifetime of n periods when the firm receives them. We say a product is at age j if it has remaining lifetime of $n + L - j$ periods. In the rest of the paper, we simply use products with lifetime i to denote products that have remaining lifetime of i periods and we only use this terminology (remaining lifetime).

Firm faces two classes of demand in each period. The first class only accepts products with lifetime longer than or equal to k, which we call as Class k demand, while the second class accepts products of any lifetime, which we call as Class 1 demand. We denote Class k demand in period t as D_t^k, Class 1 demand as D_t^1, and their realizations as d_t^k and d_t^1, respectively. $\{\mathbf{D}_t\}_{t=1,\ldots,T}$ is a sequence of i.i.d. random vectors with joint density ϕ and $\mathbf{D}_t = (D_t^k, D_t^1)$. However, demand of different classes within a period can be correlated. The decisions of how many fresh products to order (possibly none) are made at the beginning of each period before demand realization. Without loss of generality, we assume that the ordering cost per unit is 0. We also assume that there is no ordering setup cost.

Whenever there is on-hand inventory, the firm must decide whether to fulfill each unit of demand, and if so, units of which lifetime to fulfill it with. If the firm decides not to fulfill some

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demand or there is no on-hand inventory, this demand is backlogged at a penalty cost l_j per unit of Class j demand, for $j \in \{1, k\}$. We assume that $l_k \geq l_1 > 0$, i.e., Class k customers have higher priority than Class 1 customers.

At the end of each period, products with lifetime 1 is outdated and disposed. The firm also needs to decide whether to dispose products of different lifetime, and if so, how many units to dispose. For each product disposed, the firm incurs a disposal cost $\delta > 0$. The remaining products are carried to the next period with their lifetimes decreasing by one and a holding cost $h > 0$ per unit is incurred. The one-period discount factor is $\beta \in (0, 1]$. All costs are assumed to be accounted for at the beginning of each period.

The sequence of events in period t is as follows.

1. At the beginning of the period, the firm receives products that have been ordered L periods ago. Based on the on-hand inventory and the pipeline inventory, the firm places an order for fresh products (products with lifetime $n + L$), which will arrive in L periods.

2. Demands are realized and the firm decides on allocating products of different lifetime to fulfill the demands.

3. Unsatisfied demand is backlogged. The remaining products with lifetime 1 are disposed.

4. The firm decides how many products of each lifetime to dispose. The remaining products are carried to the next period with their lifetime decreasing by one.

We denote the inventory level of products with lifetime i at the beginning of period t (prior to ordering) by $\hat{x}_{i,t}$. Note that $(\hat{x}_{n+L-1,t}, \ldots, \hat{x}_{n+1,t})$ are pipeline inventory levels and $\hat{x}_{n+i,t}$ will arrive in i periods for $i = 1, \ldots, L-1$. The on-hand inventory is given by $(\hat{x}_{n,t}, \ldots, \hat{x}_{1,t})$. The order quantity in period t, y_t, can be thought of as $\hat{x}_{n+L,t}$ and it will arrive in L periods.

We denote the number of backorders of Class j demand at the beginning of period t by $\hat{b}_t^j$, for $j \in \{1, k\}$. Given the realized demand (d_t^k, d_t^1) in period t, the firm uses $a_{i,t}^j$ units of products with lifetime i to satisfy the Class j demand, for $i = 1, \ldots, n$, $j \in \{1, k\}$. Then, the on-hand inventory becomes $(\hat{x}_{n,t} - \sum_{j \in \{1,k\}} a_{n,t}^j, \ldots, \hat{x}_{1,t} - \sum_{j \in \{1,k\}} a_{1,t}^j)$. Clearly, $0 \leq \sum_{j \in \{1,k\}} a_{i,t}^j \leq \hat{x}_{i,t}$, for $i = 1, \ldots, n$, $\sum_{i=1}^{n} a_{i,t}^j \leq \hat{b}_t^j + d_t^j$ for $j \in \{1, k\}$, and $a_{i,t}^k = 0$ for $i = 1, \ldots, k-1$. Let $\hat{z}_{i,t} = \hat{x}_{i,t} - \sum_{j \in \{1,k\}} a_{i,t}^j$ for $i = 1, \ldots, n$, i.e., $\hat{z}_{i,t}$ is the inventory level of products with lifetime i after demand fulfillment. At the end of the period, $\hat{z}_{1,t}$ units of products with lifetime 1 will be disposed and $\hat{e}_t^j = \hat{b}_t^j + d_t^j - \sum_{i=1}^{n} a_{i,t}^j$

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Table 1: Notations

	System Parameters
h:	holding cost
δ:	disposal cost
l_j:	backorder cost of Class j demand
d_t^j:	realization of Class j demand in period t
β:	discount factor
	State Variables
$\hat{x}_{i,t}$:	inventory level of products with lifetime i at the beginning of period t
$\hat{b}_t^j$:	number of backorders of Class j demand at the beginning of period t
$\hat{z}_{i,t}$:	inventory level of products with lifetime i after demand fulfillment in period t
$\hat{e}_t^j$:	number of backorders of Class j demand after demand fulfillment in period t
$x_{i,t}$:	prior net inventory level of products with lifetime i in period t
$z_{i,t}$:	posterior net inventory level of products with lifetime i in period t
	Decision Variables
y_t:	order quantity for fresh products in period t
$a_{i,t}^j$:	number of products with lifetime i to satisfy the Class j demand in period t
$\hat{r}_{i,t}$:	number of products with lifetime i that are carried to the next period in period t
$r_{i,t}$:	$\hat{r}_{i,t} - z_{i,t}^-$

units of Class j demand will be backlogged, for $j \in \{1, k\}$.

The firm also needs to decide how many products of each lifetime to dispose before the end of the period. Let $\hat{r}_{i,t}$ be the number of products with lifetime i that are carried to the next period as products with lifetime $i-1$. Then, $0 \le \hat{r}_{i,t} \le \hat{z}_{i,t}$ for $i = 2, \ldots, n$. We summarize the notations in Table 1.

We use $(\hat{x}_{n+L-1,t}, \ldots, \hat{x}_{1,t}, \hat{b}_t^k, \hat{b}_t^1)$ to describe the state of the system in period t. The state is then updated as follows:

$$
\begin{aligned}
\hat{x}_{n+L-1,t+1} &= y_t; \\
\hat{x}_{n+i,t+1} &= \hat{x}_{n+i+1,t}, && \text{for } i = 0, \ldots, L-2; \\
\hat{x}_{i,t+1} &= \hat{r}_{i+1,t}, && \text{for } i = 1, \ldots, n-1; \\
\hat{b}_{t+1}^j &= \hat{b}_t^j + d_t^j - \sum_{i=1}^{n} a_{i,t}^j, && \text{for } j = 1, k.
\end{aligned}
$$

For $t = 1, \ldots, T+L$, let $\hat{v}_t(\hat{x}_{n+L-1}, \ldots, \hat{x}_1, \hat{b}^k, \hat{b}^1)$ be the optimal expected total cost from period t onward when the state of the system is $(\hat{x}_{n+L-1}, \ldots, \hat{x}_1, \hat{b}^k, \hat{b}^1)$. Then, the optimality equations

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are given by

$$\hat{v}_t(\hat{x}_{n+L-1},\ldots,\hat{x}_1,\hat{b}^k,\hat{b}^1)=\min_{y\geq 0} E\hat{w}_t(y,\hat{x}_{n+L-1},\ldots,\hat{x}_1,\hat{b}^k,\hat{b}^1,D_t^k,D_t^1), \tag{1}$$

$$\hat{w}_t(y,\hat{x}_{n+L-1},\ldots,\hat{x}_1,\hat{b}^k,\hat{b}^1,d^k,d^1)=\min_{\substack{\mathbf{0}\leq \mathbf{a}^k+\mathbf{a}^1\leq \hat{\mathbf{x}}\\ \sum_{i=1}^n a_i^j\leq \hat{b}^j+d^j, j\in\{1,k\}\\ a_i^k=0, i=1,\ldots,k-1\\ a_i^j\geq 0, i=1,\ldots,n, j\in\{1,k\}}} \Big\{\delta(\hat{x}_1-\sum_{j\in\{1,k\}} a_1^j)+\sum_{j\in\{1,k\}} l_j(\hat{b}^j+d^j-\sum_{i=1}^n a_i^j)$$
$$+\hat{u}_t(y,\hat{x}_{n+L-1},\ldots,\hat{x}_{n+1},\hat{x}_n-\sum_{j\in\{1,k\}} a_n^j,\ldots,\hat{x}_2-\sum_{j\in\{1,k\}} a_2^j,\hat{b}^k+d^k-\sum_{i=1}^n a_i^k,\hat{b}^1+d^1-\sum_{i=1}^n a_i^1)\Big\}, \tag{2}$$

and

$$\hat{u}_t(y,\hat{x}_{n+L-1},\ldots,\hat{x}_{n+1},\hat{\mathbf{z}},\hat{e}^k,\hat{e}^1)=\min_{\mathbf{0}\leq\hat{\mathbf{r}}\leq\hat{\mathbf{z}}}\{h\sum_{i=2}^n \hat{r}_i+\delta\sum_{i=2}^n(\hat{z}_i-\hat{r}_i)+\beta\hat{v}_{t+1}(y,\hat{x}_{n+L-1},\ldots,\hat{x}_{n+1},\hat{\mathbf{r}},\hat{e}^k,\hat{e}^1)\}, \tag{3}$$

where $\hat{\mathbf{x}}=(\hat{x}_n,\ldots,\hat{x}_1)$, $\mathbf{a}^j=(a_n^j,\ldots,a_1^j)$ for $j\in\{1,k\}$, $\hat{\mathbf{z}}=(\hat{z}_n,\ldots,\hat{z}_2)$, $\hat{\mathbf{r}}=(\hat{r}_n,\ldots,\hat{r}_2)$, and $\mathbf{0}=(0,\ldots,0)$ (the dimension of $\mathbf{0}$ can be identified in the context). The terminal value is assumed to be $\hat{v}_{T+L+1}(x_{n+L-1},\ldots,x_1)=0$.

The ordering, allocation, and disposal decisions are associated with (1), (2), and (3), respectively.

Lemma 1. $\hat{v}_t(\hat{x}_{n+L-1},\ldots,\hat{x}_k+\epsilon_k,\ldots,\hat{x}_1+\epsilon_1,\hat{b}^k+\epsilon_k,\hat{b}^1+\epsilon_1)=\hat{v}_t(\hat{x}_{n+L-1},\ldots,\hat{x}_1,\hat{b}^k,\hat{b}^1)$ *for all* $\epsilon_k,\epsilon_1\geq 0$.

The above result is intuitive. At the beginning of each period, if there are both products of lifetime j and backorders of Class j demand, it is optimal to use those products to fulfill the backorders as much as possible, for $j=1,k$. Note also that if $k\neq n$, it is never optimal to have products with lifetime j or $j+1$ and backorders of Class j demand at the same time after demand fulfillment, for $j=1,k$. Therefore, we can use net inventory levels to describe the state of the system. In particular, we define $x_{i,t}=\hat{x}_{i,t}-\hat{b}_t^i$ for $i=1,\ldots,n+L-1$ (define $\hat{b}_t^i=0$ when $i\neq k,1$), to which we refer as the prior net inventory level in period t, and define $z_{i,t}=\hat{z}_{i,t}-\hat{e}_t^{i-1}$ for $i=2,\ldots,n$ (define $\hat{e}_t^i=0$ when $i\neq 1,k$), to which we refer as the posterior net inventory

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level in period t. Then, $\hat{x}_{j,t} = x^+_{j,t}$, $\hat{b}_{j,t} = x^-_{j,t}$, $\hat{z}_{j+1,t} = z^+_{j+1,t}$, and $\hat{e}^j_t = z^-_{j+1,t}$, for $j = 1, k$, where $x^+ = \max\{0, x\}$ and $x^- = \max\{-x, 0\}$. Note that products perish whereas demand does not. Thus, we define both prior and posterior net inventory levels so that the posterior net inventory level after disposal in period t corresponds to the prior net inventory level in period $t+1$. Let $r_i = \hat{r}_{i,t} - z^-_{i,t}$ for $i = 2, \ldots, n$. Then, $\min\{0, z_{i,t}\} \le r_{i,t} \le z_{i,t}$ and the new state is updated as follows:

$$x_{n+L-1,t+1} = y_t; \quad x_{i,t+1} = x_{i+1,t}, \text{ for } i = n, \ldots, n+L-2; \quad x_{i,t+1} = r_{i+1,t}, \text{ for } i = 1, \ldots, n-1.$$

Let v_t and u_t be functions on $\mathbb{R}^{n+L-1-k}_+ \times \mathbb{R} \times \mathbb{R}^{k-2}_+ \times \mathbb{R}$ and w_t be a function on $\mathbb{R}^{n+L-k}_+ \times \mathbb{R} \times \mathbb{R}^{k-2}_+ \times \mathbb{R} \times \mathbb{R}^2_+$ such that

$$\begin{aligned}
v_t(x_{n+L-1}, \ldots, x_1) &= \hat{v}_t(x_{n+L-1}, \ldots, x_{k+1}, x^+_k, x_{k-1}, \ldots, x_2, x^+_1, x^-_k, x^-_1), \\
w_t(x_{n+L}, \ldots, x_1, d^k, d^1) &= \hat{w}_t(x_{n+L}, \ldots, x_{k+1}, x^+_k, x_{k-1}, \ldots, x_2, x^+_1, x^-_k, x^-_1, d^k, d^1), \\
u_t(x_{n+L}, \ldots, x_n, z_n, \ldots, z_2) &= \hat{u}_t(x_{n+L}, \ldots, x_{n+1}, z_n, \ldots, z_{k+2}, z^+_{k+1}, z_k, \ldots, z_3, z^+_2, z^-_{k+1}, z^-_2).
\end{aligned}$$

We can then rewrite the optimality equations (1), (2), and (3) as follows:

$$v_t(x_{n+L-1}, \ldots, x_1) = \min_{y \ge 0} V_t(y, x_{n+L-1}, \ldots, x_1), \tag{4}$$

$$w_t(x_{n+L}, \ldots, x_1, d^k, d^1) = \min_{\substack{\mathbf{0} \le \mathbf{a}^k + \mathbf{a}^1 \le \mathbf{x}^+ \\ \sum_{i=1}^n a^j_i \le d^j + x^-_j, j \in \{1,k\} \\ a^k_i = 0, i = 1, \ldots, k-1 \\ a^j_i \ge 0, i = 1, \ldots, n, j \in \{1,k\}}} W_t(x_{n+L}, \ldots, x_{n+1}, \mathbf{x}, d^k, d^1, \mathbf{a}^k, \mathbf{a}^1), \tag{5}$$

$$u_t(x_{n+L}, x_{n+L-1}, \ldots, x_{n+1}, \mathbf{z}) = \delta \sum_{i=2}^n z_i + \min_{\min\{\mathbf{0}, \mathbf{z}\} \le \mathbf{r} \le \mathbf{z}} U_t(x_{n+L}, \ldots, x_{n+1}, \mathbf{r}). \tag{6}$$

where

$$\begin{aligned}
V_t(x_{n+L}, x_{n+L-1}, \ldots, x_1) &= E w_t(x_{n+L}, x_{n+L-1}, \ldots, x_1, D^k_t, D^1_t), \\
W_t(x_{n+L}, \ldots, x_{n+1}, \mathbf{x}, d^k, d^1, \mathbf{a}^k, \mathbf{a}^1) &= \delta(x^+_1 - \sum_{j \in \{1,k\}} a^j_1) + \sum_{j \in \{1,k\}} l_j(x^-_j + d^j - \sum_{i=1}^n a^j_i) \\
&+ u_t(y, x_{n+L-1}, \ldots, x_{n+1}, x^+_n - \sum_{j \in \{1,k\}} a^j_n, \ldots, x^+_{k+1} - \sum_{j \in \{1,k\}} a^j_{k+1} - \hat{e}^k, \ldots, x^+_2 - \sum_{j \in \{1,k\}} a^j_2 - \hat{e}^1), \\
U_t(x_{n+L}, \ldots, x_{n+1}, \mathbf{r}) &= h \sum_{i=2}^n r^+_i - \delta \sum_{i=2}^n r_i + \beta v_{t+1}(x_{n+L}, x_{n+L-1}, \ldots, x_{n+1}, \mathbf{r}),
\end{aligned}$$

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$\hat{e}^j = x_j^- + d^j - \sum_{i=1}^{n} a_i^j$ for $j = 1, k$, and the terminal value is $v_{T+L+1}(x_{n+L-1}, \ldots, x_1) = 0$.

4 Structure of the Optimal Policy

In this section, we solve the optimality equations (4), (5), and (6) to characterize the optimal ordering, allocation, and disposal policies. For each $t = 1, \ldots, T + L$, we assume some functional properties of v_{t+1}; utilize these properties to characterize the optimal ordering, allocation, and disposal policies in period t; and then use this structure of the optimal solution to show that the assumed properties hold for v_t. We include the detailed analysis and all the proofs in the appendix. Below we state properties of the optimal cost function.

Proposition 1. *For $t = 1, \ldots, T + L + 1$, $v_t(x_{n+L-1}, \ldots, x_1)$ has the following properties:*

(1) $v_t(x_{n+L-1}, \ldots, x_1)$ is multimodular in $(x_{n+L-1}, \ldots, x_1)$,

(2) $\frac{\partial v_t}{\partial x_n} \leq \frac{\partial v_t}{\partial x_{n-1}} \leq \cdots \leq \frac{\partial v_t}{\partial x_1} \leq \delta$,

(3) $\frac{\partial v_t}{\partial x_n} \geq \frac{\partial v_t}{\partial x_k} - l_k$ and $\frac{\partial v_t}{\partial x_{k-1}} \geq \frac{\partial v_t}{\partial x_1} - l_1$.

Note that multimodularity implies convexity and supermodularity (see Appendix A for the definition of multimololularity and its properties). Thus property (1) of Proposition 1 implies that $v_t(x_{n+L-1}, \ldots, x_1)$ is jointly convex and supermodular. Property (2) shows that having one more fresher product on hand is always better than having one more older product. This means that for on-hand inventory, a fresher unit is more valuable than an older one. Note that this is not always true for the pipeline inventory. Products that will arrive earlier can be used to fulfill demand earlier and thus may have larger values. However, those products also have a shorter lifetime at any given time. Therefore, for the pipeline inventory, the value of each product depends on both the lifetime and the leadtime. Property (2) also shows that having one more backorder of Class 1 demand is better than having one more backorder of Class k demand. This is because Class k demand requires products with lifetime longer than or equal to k whereas Class 1 demand accepts products of any age. Property (3) shows that difference between the benefit of having one more unit of the product with lifetime n $(k-1)$ and that of having one more unit of the product with lifetime k (1) (or that of having one less backorder of Class k (1) demand) is at most l_k (l_1).

In the following, we first consider optimality equation (6) to characterize the structure of the

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optimal disposal policy. Note that $(x_{n+L,t}, x_{n+L-1,t}, \ldots, x_{n+1,t}, z_{n,t}, \ldots, z_{1,t})$ is the posterior net inventory levels after demand fulfillment in period t. For $i = 2, \ldots, n$, let

$$r^*_{i,t} = \max \operatorname*{argmin}_{r \geq 0} U_t(x_{n+L}, \ldots, x_{n+1}, z_n, \ldots, z_{i+1}, r, 0, \ldots, 0). \tag{7}$$

We show that the optimal disposal policy can be characterized by these $n-1$ thresholds in the following theorem.

Theorem 1 (Optimal Disposal Policy)**.** *In each period, it is optimal to keep* $\min\{z_{i,t}, r^*_{i,t}\}$ *units of products with lifetime* i *for each* $i = 2, \ldots, n$ *and dispose the rest of the products, where* $r^*_{i,t}$ *is defined by (7). Moreover, the optimal thresholds* $r^*_{i,t}$ *have the following properties:*
(1) $r^*_{i,t}$ *only depends on* $(x_{n+L,t}, x_{n+L-1,t}, \ldots, x_{n+1,t}, z_{n,t}, \ldots, z_{i+1,t})$*;*
(2) If $r^*_{i,t} < z_{i,t}$*, then* $r^*_{j,t} = 0$ *for all* $j < i$*.*

Theorem 1 shows that the optimal disposal policy can be characterized by $n-1$ thresholds, where each threshold specifies the reservation level for products of a particular lifetime. Property (1) states that each reservation level only depends on the inventory levels of products with longer lifetime. Property (2) states that if it is optimal to dispose products of age i, then it is also optimal to dispose all older products. In other words, it is optimal to keep products with lifetime i only when there are not enough younger products. Therefore, it is optimal to dispose products with lifetime 2 first, then only once there are no products of lifetime 2, disposing products of lifetime 3 should be considered, and so on.

In the following, we identify more properties of the optimal thresholds.

Proposition 2. *For* $t = 1, \ldots, T$*, the optimal thresholds* $r^*_{i,t}$ *have the following properties.*
(1) For $i = 2, \ldots, n-1$*,* $r^*_{i,t} \leq \max\{r^*_{i+1,t} - z_{i+1,t}, 0\}$*.*
(2) For $i = 2, \ldots, n$*,* $r^*_{i,t}$ *is nonincreasing in* $z_{j,t}$ *and* $x_{l,t}$ *for* $j = i+1, \ldots, n$ *and* $l = n+1, \ldots, n+L$*, and* $-1 \leq \frac{\partial r^*_{i,t}}{\partial z_{i+1,t}} \leq \cdots \leq \frac{\partial r^*_{i,t}}{\partial z_{n,t}} \leq \frac{\partial r^*_{i,t}}{\partial x_{n+1,t}} \leq \cdots \leq \frac{\partial r^*_{i,t}}{\partial x_{n+L,t}} \leq 0$*.*

Property (1) of Proposition 2 shows that $r^*_{i,t}$ is bounded from above by the difference between $r^*_{i+1,t}$ and $z_{i+1,t}$. When there are many fresher products (with lifetime longer than i) remaining, i.e., $z_{i+1,t} \geq r^*_{i+1,t}$, the firm does not need to keep older products ($r^*_{i,t} = 0$). When there are not enough fresh products remaining, i.e., $z_{i+1,t} < r^*_{i+1,t}$, the firm should keep at most $r^*_{i+1,t} - z_{i+1,t}$

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units of the older products. This also implies that the total number of products that are carried to the next period should be no larger than $r^*_{n,t}$.

Property (2) shows that the optimal reservation threshold for products of a particular lifetime would decrease if there are more fresher products. Among those fresher products, this threshold is more sensitive to the inventory of the older product. However, the impact of inventory on the thresholds is limited. The decreasing rate of the thresholds with respect to the inventory of other products is at most 1.

Next, we consider optimality equation (5) to characterize the structure of the optimal allocation policy. Note that Class k demand has higher priority than Class 1 demand since it has a higher backorder cost. Therefore, it is optimal to fulfill Class k demand first.

Lemma 2. *In each period, it is optimal to satisfy Class k demand before Class 1 demand. When fulfilling each demand class, it is optimal to use products with lifetime i before products with lifetime j for $i < j$.*

Let $(\tilde{x}_{n+L,t}, \ldots, \tilde{x}_{1,t})$ be the posterior net inventory levels after satisfying Class k demand. Let $\tilde{d}^1_{i,t} = (d^1_t + \tilde{x}^-_{1,t} - \sum_{j=1}^{i-1} \tilde{x}^+_{j,t})^+$ for $i = k+1, \ldots, n$, which denotes the remaining Class 1 demand after using products with lifetime shorter than i to satisfy Class 1 demand as much as possible. It is optimal to use products with lifetime i to fulfill Class 1 demand only if $\tilde{d}^1_{i,t} > 0$. We also define $\tilde{b}^1_{i,t} = \tilde{d}^1_{i,t} - \tilde{x}_{i,t}$ as the difference between the remaining Class 1 demand and the inventory level of products with lifetime i. For $i = k+1, \ldots, n$, let

$$\eta^*_{i,t} = \max \operatorname*{argmin}_{\eta \geq 0} \{u_t(\tilde{x}_{n+L}, \ldots, \tilde{x}_{i+1}, \eta, 0, \ldots, 0, -\tilde{b}^1_i - \eta) + l_1 \eta\}. \tag{8}$$

In the following theorem, we show that the optimal allocation policy can be characterized by these $n-k$ thresholds.

Theorem 2 (Optimal Allocation Policy)**.** *In period t, the optimal allocation policy can be described as follows:*

Step 1*: It is optimal to first satisfy Class k demand as much as possible using products with lifetime longer than or equal to k following the FIFO policy.*

Step 2*: It is optimal to reserve $\min\{\tilde{x}_{i,t}, \eta^*_{i,t}\}$ units of products with lifetime i, for $i = k+1, \ldots, n$, and use the rest to fulfill Class 1 demand as much as possible following the FIFO policy, where $\eta^*_{i,t}$*

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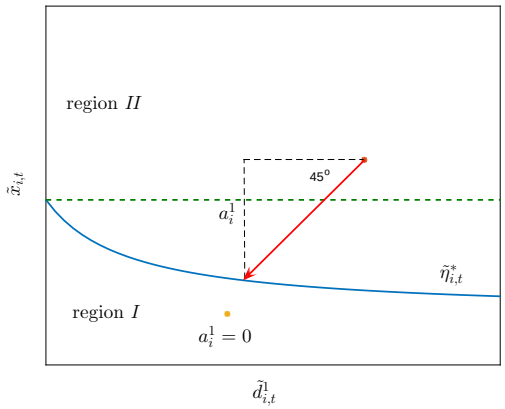


Figure 1: Illustration of the optimal allocation policy

is defined by (8). Moreover, the optimal threshold $\eta_{i,t}^*$ *has the following properties:*

(1) $\eta_{i,t}^*$ *only depends on* $(\tilde{x}_{n+L,t}, \ldots, \tilde{x}_{i+1,t})$ *and* $\tilde{b}_{i,t}^1$*;*

(2) if $\eta_{i,t}^* < \tilde{x}_{i,t}$*, then* $\eta_{j,t}^* = 0$ *for all* $j < i$*.*

Theorem 2 shows that it is optimal to fulfill Class k demand first. This is because Class k demand has higher priority than Class 1 demand since it has a higher backorder cost. When fulfilling each demand class, it is optimal for the firm to use the FIFO allocation policy. However, when fulfilling Class 1 demand, the firm may need to reserve fresher products with lifetime $k+1$ or higher for future Class k demand. When $k = n$, the firm does not need to reserve any products. Property (2) states that if it is optimal not to reserve all products with lifetime i, then it is also optimal not to reserve all older products. In other words, it is optimal to reserve products with lifetime i only when there are not enough fresher products. Therefore, it is optimal to first reserve products with lifetime n, then to reserve products with lifetime $n-1$ if needed and so on.

Figure 1 illustrates how to use products with lifetime i to satisfy Class 1 demand given $\tilde{x}_{i,t}$ and $\tilde{d}_{i,t}^1$. As we can see, the entire space is divided by the optimal rationing thresholds into two regions. In region *I*, it is optimal to do nothing. That is, it is optimal not to use products with lifetime i to fulfill Class 1 demand. In region *II*, it is optimal to reduce both $\tilde{x}_{i,t}$ and $\tilde{d}_{i,t}^1$ along the red arrow until it reaches either the rationing threshold or the y axis. That is, it is optimal to reserve $\eta_{i,t}^*$ units of products with lifetime i and use the rest to fulfill demand as much as possible. Moreover, we can see that for points that are on the same 45 degree line (those points have the same difference between $\tilde{x}_{i,t}$ and $\tilde{d}_{i,t}^1$), the optimal rationing threshold is the same. Thus, the optimal rationing

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threshold depends only on the difference of $\tilde{x}_{i,t}$ and $\tilde{d}^1_{i,t}$, or equivalently, on $\tilde{b}^1_{i,t}$.

Figure 1 also shows that the optimal rationing threshold is decreasing in $\tilde{d}^1_{i,t}$ for fixed $\tilde{x}_{i,t}$ (we analytically prove this in Proposition 3). This suggests that the optimal rationing threshold would decrease if there are more Class 1 demand. This differs from the existing results in the literature on inventory rationing. These papers either assume there is no ordering decision or show that the base stock ordering policy is optimal. Thus, the only tradeo for rationing is to balance the backorder cost of demand with lower priority plus the holding cost with (possible) future backorder cost of demand with higher priority. However, in our model, since products are perishable, products with different lifetime have different values. As a consequence, the optimal order quantity is nonlinear in the net inventory of products with different lifetime (see Theorem 3 below). For example, if one unit of fresh products is used to fulfill one backorders of demand with lower priority, then the optimal order quantity would be larger in the next period. Therefore, when deciding on how many fresh products to reserve, the firm needs to take into account how the inventory of fresh products and the number of backorders would affect the ordering decision. It is worth to mention that if a base stock ordering policy is applied, then we can show that the optimal rationing level $\eta^*_{i,t}$ is independent of $\tilde{b}^1_{i,t}$ and thus independent of d^1_t and $\tilde{x}_{m,t}$ for $m = 1, \ldots, i$ (the threshold becomes the green dotted line in Figure 1).

Different from most inventory rationing literature in which rationing in due to limited capacity, in our model, the reason for the firm to reserve fresher products for future Class k demand is that there is uncertainty about the leadtime demand. Since the order placed in period t will arrive in period $t + L$, this order quantity might not be enough to fulfill Class k demand in period $t + L$ if the realized demand during the leadtime (from period t to period $t + L - 1$) is large. Thus, the firm may need to reserve fresher products to fulfill Class k demand in period $t + L$. If there is no uncertainty about the leadtime demand, for example, if the demand in period τ, $\tau = t, \ldots, t+L-1$, is deterministic, the optimal order quantity in period t will cover demand in period $t + L$ (since the on-hand inventory levels at the end of period $t + L - 1$ are deterministic given the on-hand inventory levels and the pipeline inventory levels in period t). In this case, the firm need not reserve any products. Similarly, if the leadtime is 0, the firm need not reserve any products either.

Our assumption that the firm can intentionally dispose products of any lifetime allows the firm to reduce the holding cost through disposal when there is excess on-hand inventory. Therefore, a

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fresher unit on hand is always more valuable than an older one. Hence, when fulfilling each demand class, it is optimal to use FIFO policy until the on-hand inventory reaches the rationing threshold. If inventory cannot be intentionally disposed, i.e., only products with lifetime 1 can be disposed, Lemma 2 still holds. It is optimal to satisfy Class 1 demand before Class 2 demand. However, when fulfilling each demand class, as mentioned in Chen et al. (2014), FIFO allocation policy may not be optimal. In this case, charactering the structure of the optimal allocation policy is difficult since one cannot easily identify which products are less valuable than others.

The rationing thresholds $\eta^*_{i,t}$ have similar properties as the reservation thresholds $r^*_{i,t}$. We state them in the following proposition.

Proposition 3. *For $t = 1, \ldots, T$, the rationing thresholds $\eta^*_{i,t}$ have the following properties.*
(1) For $i = k+1, \ldots, n-1$, $\eta^*_{i,t} \leq \max\{\eta^*_{i+1,t} - \tilde{x}_{i+1,t}, 0\}$.
(2) For $i = k+1, \ldots, n$, $\eta^*_{i,t}$ is nonincreasing in $\tilde{x}_{j,t}$ and $\tilde{b}^1_{i,t}$ for $j = i+1, \ldots, n+L$, with $-1 \leq \frac{\partial \eta^*_{i,t}}{\partial \tilde{b}^1_{i,t}} \leq 0$ and $-1 \leq \frac{\partial \eta^*_{i,t}}{\partial \tilde{x}_{i+1,t}} \leq \cdots \leq \frac{\partial \eta^*_{i,t}}{\partial \tilde{x}_{n+L,t}} \leq 0$.

Property (1) in Proposition 3 shows that $\eta^*_{i,t}$ is bounded from above by the difference between $\eta^*_{i+1,t}$ and $\tilde{x}_{i+1,t}$. When there are many fresher products (with lifetime longer than i), i.e., $\tilde{x}_{i+1,t} \geq \eta^*_{i+1,t}$, the firm does not reserve any older products for future Class k demand ($\eta^*_{i,t} = 0$). When there are not enough fresh products remaining, i.e., $\tilde{x}_{i+1,t} < \eta^*_{i+1,t}$, the firm should reserve at most $\eta^*_{i+1,t} - \tilde{x}_{i+1,t}$ units of the older products. This also implies that the total number of products reserved for future Class k should be no larger than $\eta^*_{n,t}$.

Property (2) shows that the optimal rationing threshold would decrease if there are more fresher products. Among those fresher products, this threshold is more sensitive to the inventory of the older products. However, the impact of inventory on the thresholds is limited. The decreasing rate of the thresholds with respect to the inventory of other products is at most 1. Property (2) also suggests that the optimal rationing threshold is decreasing in the difference between the remaining Class 1 demand and the inventory level with decreasing rate being less than or equal to 1.

Note that the optimal allocation policy for Class 1 demand has similar structure as the optimal disposal policy and the rationing level $\eta^*_{i,t}$ has similar properties to $r^*_{i,t}$. This is because one may view the allocation problem as a two-stage problem where the firm first decides on how to fulfill Class k demand and then decides on how to fulfill Class 1 demand since it is optimal to satisfy

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Class k demand before Class 1 demand. Fulfilling the Class 1 demand is also similar to a clearance problem since the backorder cost for Class 1 demand is lower. However, in contrast to the clearance problem, in our model demand for products on sale (D^1) is finite and random rather than infinite, unfulfilled demand is backlogged rather than lost, and we allow a positive lead time (see, e.g., Li and Yu 2014).

In the following proposition, we show that the optimal thresholds in the allocation policy are smaller than or equal to the optimal thresholds in the disposal policy.

Proposition 4. *$\eta^*_{i,t} \leq r^*_{i,t}$ for $i = k+1, \ldots, n-1$.*

This result is intuitive. It suggests that it is never optimal to dispose any product in a period in which there are backorders of Class 1 demand.

Next, we examine the optimal ordering policy.

Theorem 3 (Optimal Ordering Policy)**.** *The optimal order quantity y^*_t is nonincreasing in x_i for $i = 1, \ldots, n+L-1$. Moreover, $-1 \leq \frac{\partial y^*_t}{\partial x_{n+L-1}} \leq \cdots \leq \frac{\partial y^*_t}{\partial x_1} \leq 0$.*

Theorem 3 shows that the optimal order quantity is nonincreasing in the net inventory levels of products with different lifetimes and it is more sensitive to the net inventory of the fresher products. The decreasing rate of the optimal order quantity with respect to the net inventory levels is at most 1. Similar result appears in Nahmias (1975), Fries (1975), Chen et al. (2014) and Li and Yu (2014) under different settings.

5 Model with $m(\leq n)$ Demand Classes

In this section, we extend the original model to include general $m(\leq n)$ demand classes. Each class only accepts products that have lifetime longer than or equal to a certain periods. Without loss of generality, we assume there are n demand classes (the model with m demand classes can be treated as a special case by letting demand of some classes be 0). The jth demand class only accepts products with lifetime longer than or equal to j, which we call as Class j demand, for $j = 1, \ldots, n$. We denote Class j demand in period t as D^j_t and its realization as d^j_t. $\{\mathbf{D}_t\}_{t=1,\ldots,T}$ is a sequence of i.i.d. random vectors with joint density ϕ_n and $\mathbf{D}_t = (D^n_t, \ldots, D^1_t)$. However, demand of different classes within a period can be correlated. The backorder cost for Class j demand is l_j. We assume

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that $0 < l_1 \leq l_2 \leq \cdots \leq l_n$, i.e., Class i customers have higher priority than Class j customers for $i \geq j$. Other assumptions are the same as those of the model with two demand classes.

Let $\hat{b}_t^j$ be the number of backorders of Class j demand at the beginning of period t and $\hat{e}_t^j$ be be the number of backorders of Class j demand after demand fulfillment in period t. Similar to the model with two demand classes, we define the prior net inventory levels as $x_{i,t} = \hat{x}_{i,t} - \hat{b}_t^i$ for $i = 1, \ldots, n$ and define the posterior net inventory levels as $z_{i,t} = \hat{z}_{i,t} - \hat{e}_t^{i-1}$ for $i = 2, \ldots, n+1$. Let $a_{i,t}^j$ be the number of products with lifetime i that are used to satisfy the Class j demand in period t, for $i = 1, \ldots, n$ and $j = 1, \ldots, n$. Other notations are the same as those of the model with two demand classes. Then, the optimality equations are given by

$$v_t(x_{n+L-1}, \ldots, x_1) = \min_{y \geq 0} E w_t(y, x_{n+L-1}, \ldots, x_1, \mathbf{D}_t), \tag{9}$$

$$\begin{aligned} w_t(y, x_{n+L-1}, \ldots, x_1, \mathbf{d}) = \min_{\substack{\mathbf{0} \leq \sum_{j=1}^n \mathbf{a}^j \leq \mathbf{x}^+ \\ \sum_{i=1}^n a_i^j \leq d^j + x_j^-, j=1,\ldots,n \\ a_i^j = 0, i<j \\ a_i^j \geq 0, i=1,\ldots,n, j=1,\ldots,n}} & \Bigg\{ \delta(x_1^+ - \sum_{j=1}^n a_1^j) + \sum_{j=1}^n l_j (d^j + x_j^- - \sum_{i=1}^n a_i^j) \\ & + u_t(y, x_{n+L-1}, \ldots, x_{n+1} - \hat{e}^n, x_n^+ - \sum_{j=1}^n a_n^j - \hat{e}^{n-1}, \ldots, x_2^+ - \sum_{j=1}^n a_2^j - \hat{e}^1) \Bigg\}, \end{aligned} \tag{10}$$

and

$$u_t(y, x_{n+L-1}, \ldots, x_{n+1}, \mathbf{z}) = \min_{\min\{\mathbf{0}, \mathbf{z}\} \leq \mathbf{r} \leq \mathbf{z}} \{ h \sum_{i=2}^n r_i^+ + \delta \sum_{i=2}^n (z_i - r_i) + \beta v_{t+1}(y, x_{n+L-1}, \ldots, x_{n+1}, \mathbf{r}) \}, \tag{11}$$

where $\mathbf{d} = (d^n, \ldots, d^1)$, $\hat{e}^j = d^j + x_j^- - \sum_{i=1}^n a_i^j$ for $j = 1, \ldots, n$, and the terminal value is 0.

Using similar analysis as in the model with two demand classes, we can show that $v_t(x_{n+L-1}, \ldots, x_1)$ has the following properties:

(1) $v_t(x_{n+L-1}, \ldots, x_1)$ is multimodular in $(x_{n+L-1}, \ldots, x_1)$, and

(2) $\frac{\partial v_t}{\partial x_n} \leq \frac{\partial v_t}{\partial x_{n-1}} \leq \cdots \leq \frac{\partial v_t}{\partial x_1} \leq \delta$.

Using these properties, we can show that the optimal disposal and ordering policies are the same as those in the model with two demand classes, which are described in Theorems 1 and 3.

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Moreover, the optimal allocation policy is described in the following theorem.

Theorem 4 (Optimal Allocation Policy)**.** *In period t, the optimal allocation policy is a sequential rationing policy:*

Step 1*: It is optimal to first satisfy Class n demand as much as possible using products with lifetime n and then satisfy Class $n-1$ demand as much as possible using products with lifetime $n-1$ and n following the FIFO policy.*

Step $j \geq 2$ (fulfilling Class $i = n-j$ demand)*: Let $(\tilde{x}^i_{n+L,t}, \ldots, \tilde{x}^i_{1,t})$ be the net inventory levels after Step $j-1$ and $\tilde{b}^i_{k,t} = (d^i_t + (\tilde{x}^i_{i,t})^- - \sum_{j=i}^{k-1} (\tilde{x}^i_{j,t})^+)^+ - (\tilde{x}^i_{k,t})^+$ for $k = i+2, \ldots, n$. Then, there exit $j-1$ thresholds $\eta^i_{k,t}$ $(k = i+2, \ldots, n)$ such that it is optimal to reserve $\min\{(\tilde{x}^i_{k,t})^+, \eta^i_{k,t}\}$ units of products with lifetime k, for $k = i+2, \ldots, n$, and use the rest of products which have lifetime longer than or equal to i to fulfill Class i demand as much as possible following the FIFO policy. Moreover, the optimal threshold $\eta^i_{k,t}$ has the following properties:*

(1) $\eta^i_{k,t}$ depends on $(\tilde{x}^i_{n+L,t}, \ldots, \tilde{x}^i_{k+1,t}, \tilde{x}^i_{i-1,t}, \ldots, \tilde{x}^i_{1,t})$, $\tilde{b}^i_{k,t}$, and $(d^{i-1}_t, \ldots, d^1_t)$;

(2) if $\eta^i_{k,t} < \tilde{x}^i_{k,t}$, then $\eta^i_{j,t} = 0$ for all $j < k$;

(3) $\eta^i_{k,t} \leq \max\{\eta^i_{k+1,t} - \tilde{x}^i_{k+1,t}, 0\}$ for $k = i+2, \ldots, n-1$;

(4) $\eta^i_{k,t}$ is nonincreasing in $(x_{n+L,t}, x_{n+L-1,t}, \ldots, x_{n+1,t}, \tilde{x}^i_{n,t}, \ldots, \tilde{x}^i_{k+1,t}, \tilde{x}^i_{i-1,t}, \ldots, \tilde{x}^i_{1,t})$ and $\tilde{b}^i_{k,t}$, and nondecreasing in $(d^{i-1}_t, \ldots, d^1_t)$. Moreover, $-1 \leq \frac{\partial \eta^i_{k,t}}{\partial \tilde{b}^i_{k,t}} \leq 0$, $-1 \leq \frac{\partial \eta^i_{k,t}}{\partial \tilde{x}^i_{k+1,t}} \leq \cdots \leq \frac{\partial \eta^i_{k,t}}{\partial \tilde{x}^i_{n+L,t}} \leq 0$, and $-1 \leq \frac{\partial \eta^i_{k,t}}{\partial \tilde{x}^i_{i-1,t}} \leq \cdots \leq \frac{\partial \eta^i_{k,t}}{\partial \tilde{x}^i_{1,t}} \leq 0$.

Theorem 4 shows that the optimal allocation policy is a sequential rationing policy: first satisfy the demand with the highest priority, then the demand with the second highest priority, and so on, such that while fulfilling each demand class, it is optimal to ration fresher products to certain thresholds for future demand with higher priority, and use the rest to fulfill demand of that class as much as possible following the FIFO policy. The rationing threshold of products with lifetime k when fulfilling Class i demand is $\eta^i_{k,t}$.

Note that the optimal rationing threshold $\eta^i_{k,t}$ depends on both the inventory levels of products with lifetime longer than k and levels of products with lifetime shorter than i. Moreover, it depends on the realized demand in classes lower than i (in the case of two demand classes, $\eta^*_{j,t} = \eta^1_{j,t}$ and thus only depends on the inventory levels of products with lifetime longer than j). In other words, when deciding on how many products of lifetime k to reserve when fulfilling Class i demand, we

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should also take into account how to allocation the rest of products to the rest of demand.

The intuition is as follows. Products with lifetime k reserved in period t can be used to fulfill Class j demand in period $t+1$, for $j = 1, \ldots, k-1$. Thus the marginal value of these products depends on the inventory levels of products with lifetime j in period $t+1$, i.e., the inventory levels of products with lifetime $j+1$ at the end of period t, for $j = 1, \ldots, k-1$, and is decreasing in the inventory levels. We will reserve one unit of products with lifetime k if and only if its marginal value is not less than $h + l_i$. Therefore, the rationing threshold $\eta^i_{k,t}$ depends on the inventory levels of products with lifetime shorter than i at the end of period t, which depends on how to allocate $(\tilde{x}^i_{k-1}, \ldots, \tilde{x}^i_1)$ to the lower demand classes. It is worth to mention that if a base stock ordering policy is applied, then we can show that the optimal rationing level for products with lifetime i only depends on the inventory levels of fresher products (with lifetime longer than i). In this case, the optimal allocation policy is the same as the parallel and sequential rationing policy in Yu et al. (2015) with $\eta^i_{i+1,t} = 0$.

In addition, the optimal rationing threshold would decrease if there are more products with lifetime longer than i or more products with lifetime shorter than j. Among products with lifetime longer than i, this threshold is more sensitive to the inventory of the older products. Among products with lifetime shorter than j, this threshold is more sensitive to the inventory of the fresher ones. However, the impact of inventory on the thresholds is limited. The decreasing rate of the thresholds with respect to the inventory of other products is at most 1.

We end this section by showing that $\eta^j_{i,t}$ is monotone in j and it is smaller than or equal to the optimal thresholds in the disposal policy.

Proposition 5. *$\eta^j_{i,t} \leq \eta^l_{i,t}$ for $1 \leq l < j < i \leq n$ and $\eta^1_{i,t} \leq r^*_{i,t}$ for $i = 2, \ldots, n$.*

This result suggests that if it is not optimal to use products of lifetime i to fulfill Class j demand, then it is also not optimal to use products of lifetime i to fulfill Class l demand, for $l < j$. Moreover, if there is unfulfilled Class 1 demand, then it is optimal not to dispose any product.

6 Managerial Implications

In this section, we provide more managerial insights into our problem. We restrict ourselves to the case with two demand classes. The results for cases with n demand classes are similar. We consider

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two important managerial questions in the control of perishable items.

1. How robust and e ective is the optimal control?

We consider this question in terms of: (i) How robust is the optimal policy we derive to other settings, including settings with an infinite planning horizon, settings with lostsales, and settings with no intentional disposal? (ii) How significant is the cost advantage of the optimal policy in comparison to a simple heuristic and what part of the control, ordering, rationing, or disposing, is more important? (iii) Does the curse of dimensionality for dynamic programming makes calculating the optimal policy too time consuming in practice?

From our results we find that (a) the structure of the optimal policy is robust to different assumptions and the optimal ordering and allocation policies are close to optimal even if the firm cannot dispose products; (b) ignoring difference between demand classes and using simple policies can significantly increase the total cost. We also develop an effective and computationally-efficient heuristic which is at most 5% away from the optimal in our numerical examples.

2. How can the firm improve the control of perishables?

We consider this question by examining the impact of the leadtime, the lifetime, and the demand class. Clearly, the firm can reduce its inventory control costs by (a) increasing the products' lifetime, (b) decreasing the products' leadtime, and (c) increasing customers' willingness to accept older products. We show that the benefit of decreasing the leadtime is more significant than the benefit of increasing the lifetime of the products and the benefit of increasing the lifetime of the products is more significant than the benefit of decreasing the acceptance threshold of the demand.

6.1 Robustness and E ectiveness of the Optimal Policy

In this subsection, we answer the first question: how robust and effective is the optimal policy? We first show that the structure of the optimal policy is robust to different assumptions.

First, our results can be extended to the case with non-stationary parameters, with any $l_1, \ldots, l_n$, and with a general terminal value which satisfies properties (1) and (2) in Section 5. Our results also hold in the setting with an infinite planning horizon under the expected total discounted cost criterion or the average cost criterion. Moreover, when the supply of perishable goods is exogenous and stochastic, our results continue to hold.

When unfulfilled demand is lost, we can use the convexity to show that the optimal allocation

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and disposal policies are the same. However, the optimal ordering policy is different from that in the backorder case. That is, the optimal order quantity may be more sensitive to the inventory of older products than that of fresher products. Similarly, when the disposal cost is age dependent, we can show that the optimal allocation policy is a sequential rationing allocation policy and the optimal disposal policy is characterized by $n-1$ thresholds. However, the optimal order quantity may be more sensitive to the inventory of older products than that of fresher products.

Next, we examine the effectiveness of the optimal policy. To do so, we first consider a benchmark model, in which a simple policy is implemented. Then, we compare the optimal cost with the cost of the benchmark model to evaluate the benefit of the optimal policy.

For fixed L, n and k, we set Class j demand to be truncated normally distributed on [0,30] with mean μ^j and standard deviation $\mu^j/3$, for $j = 1, k$. Class k demand is independent of Class 1 demand. We consider the following cases: $T \in \{5, 10, 15\}$, $\mu^j \in \{10, 15, 20\}$ for $j = 1, k$, $h \in \{1, 3, 5\}$, $l_1 = 3$, $l_k \in \{4, 7, 10\}$, and $\delta \in \{1, 3, 5\}$. "Max" and "Median" in all the following tables correspond to the maximum value and the median value in all these cases, respectively. All computations were carried out using Matlab R2015a on a computer workstation with an Intel Core i7-4790 processor.

In the benchmark model, the optimal ordering, allocation, and disposal policies are specified as follows:

Ordering policy: Myopic ordering policy with leadtime L, i.e., the order quantity y_t^{B} in state $(x_{n+L-1}, \ldots, x_1)$ is the solution to

$$P(y_t^{\mathrm{B}} + \sum_{i=1}^{n+L-1} x_{i,t} \leq D^1 + D^k) = \frac{l_s}{l_s + h},$$

where $l_s = \frac{\mu^1 l_1 + \mu^k l_k}{\mu^1 + \mu^k}$, $\mu^j = E(D^j)$ for $j = 1, k$.

Note that the ordering policy in the benchmark is the optimal ordering policy in a system with nonperishable products, single demand class and fixed leadtime L (see for example, section 9.6.2 of Zipkin 2000). Thus, the benchmark model ignores the perishability feature of the product and the difference in the demand classes (since no previous literature considers perishable inventory systems with multiple demand classes and there are no known heuristics, we came up with this reasonable simple policy).

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Table 2: Benefit of the optimal policy

			Δ^{B}		Δ^{O}		Δ^{A}		Δ^{D}	
L	n	k	Max	Median	Max	Median	Max	Median	Max	Median
1	2	2	19.2	5.7	19.2	5.7	0	0	0	0
-	3	2	21.0	11.5	12.4	4.3	18.3	7.7	0	0
-	3	3	15.9	4.8	15.9	4.8	0	0	0	0
-	4	2	21.8	11.5	10.1	4.0	21.6	8.2	0	0
-	4	3	23.1	11.6	11.9	4.3	20.8	8.0	0	0
-	4	4	15.6	4.7	15.6	4.7	0	0	0	0
2	2	2	19.0	5.9	18.8	5.9	0	0	0	0
-	3	2	27.2	12.6	13.0	4.5	23.2	8.7	0.3	0.1
-	3	3	16.5	4.9	16.4	4.9	0	0	0.2	0.1
3	2	2	18.2	5.8	18.2	5.8	0	0	0.5	0.2

Allocation policy: Sequential FIFO allocation policy, i.e., the demand class with high priority is satisfied as much as possible following the FIFO policy, and then the demand class with low priority is satisfied as much as possible following the FIFO policy using the remaining products.

Disposal policy: Minimum disposal policy, i.e., the firm only dispose products with lifetime 1 at the end of each period.

Let $v_t^{\mathrm{OPT}}(\mathbf{x})$ be the optimal expected total cost and $v_t^{\mathrm{B}}(\mathbf{x})$ be the expected total cost of the benchmark model with the initial state $\mathbf{x} = (x_{n+L-1}, \ldots, x_1)$ in period t. We define

$$\Delta^{\mathrm{B}} = \frac{v_1^{\mathrm{B}}(\mathbf{x}) - v_1^{\mathrm{OPT}}(\mathbf{x})}{v_1^{\mathrm{OPT}}(\mathbf{x})} * 100$$

as the relative difference between the cost of the benchmark model and the optimal cost.

In Table 2, we provide representative results comparing the optimal cost with the cost of the benchmark model. As we can see, the relative difference can be as large as 27%. The medians for most of the cases are above 4.5% (above 11% when $k < n$). This suggests that the optimal policy performs significantly better than the policy in the benchmark model. In other words, ignoring perishability, the difference in the demand classes, and using simple policies can lead to significant increase in the total cost.

We are also interested in the benefit of each individual component of the optimal policy (the optimal ordering policy, allocation policy, and disposal policy). To evaluate the benefit of the optimal ordering policy, we compare the optimal cost with the expected total cost if the firm uses the optimal allocation and disposal policies and the myopic ordering policy in the benchmark model.

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Let $v_t^{\mathrm{O}}(\mathbf{x})$ be the expected total cost under the myopic ordering policy and the optimal allocation and disposal policies. Similarly, to evaluate the benefit of the optimal allocation policy and the benefit of the optimal disposal policy, let $v_t^{\mathrm{A}}(\mathbf{x})$ be the expected total cost under the sequential allocation policy and the optimal ordering and disposal policies and $v_t^{\mathrm{D}}(\mathbf{x})$ be the expected total cost under the minimum disposal policy and the optimal ordering and allocation policies. We define

$$\Delta^i = \frac{v_1^i(\mathbf{x}) - v_1^{\mathrm{OPT}}(\mathbf{x})}{v_1^{\mathrm{OPT}}(\mathbf{x})} * 100, \qquad \text{for } i\text{=O, A, D.}$$

Table 2 also provides the maximum values and the medians of Δ^{O}, Δ^{A}, and Δ^{D} under different parameters, respectively.

As we can see, the benefit of the optimal ordering policy and the benefit of the optimal allocation policy can be significant. The benefit of the optimal ordering policy can be as large as 19% while the benefit of the optimal allocation policy can be as large as 23%. This suggests that applying the optimal ordering and allocation policies is important in our problem. Note that when $k = n$, it is optimal not to reserve any products for future Class k demand. In this case, the sequential allocation policy becomes optimal. However, when $k < n$, as we can see, the sequential allocation policy perform much worse than the optimal policy .

We can also see that the benefit of the optimal disposal policy is limited. This is because it is optimal to dispose some inventory that is yet to expire only when the inventory level of the products is large. However, when the initial inventory is low, these states have a low probability to occur. Thus the firm can simply dispose products with lifetime 1 at the end of each period without significantly increasing the cost. Note that for a system in which disposal is not allowed, the optimal cost is larger than the optimal cost in our model. Therefore, if our optimal ordering and allocation policies are implemented in such a system, the relative difference between the resulting cost and the optimal cost is smaller than Δ^D. This implies that for a system in which disposal is not allowed, our optimal ordering and allocation policies can perform very well.

6.2 A Simple E ective Heuristic and its Performance

We have shown that the optimal policy performs significantly better than the simple policy in the benchmark model. However, obtaining the optimal thresholds and the optimal order quantities

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based on the dynamic programming formulation can be computationally intensive. This is due to the well known curse of dimensionality associated with dynamic programming (the state space grows exponentially with the leadtime and the lifetime). Therefore, an effective and computationally-efficient heuristic is needed.

We propose a heuristic that is based upon the optimal policy and that is easy to compute:

Disposal policy: Minimum disposal (also used in the benchmark model).

We have already shown that the benefit of the optimal disposal policy is limited. Therefore, the minimum disposal policy is a good candidate as a heuristic for the optimal disposal policy.

Allocation policy: Minimum fixed rationing policy, i.e., the firm reserves up to η_t^H units of freshest products which have lifetime longer than k, where $\eta_t^H = 0$ if $L = 0$ and otherwise η_t^H equals to the solution to the following equation

$$l_1 + h = \beta(l_k - l_1)P(D_{t+1}^k \geq \eta_t^H). \tag{12}$$

In section 4, we show that the optimal allocation policy is a sequential rationing policy which is characterized by different thresholds. These thresholds are state-dependent and difficult to compute. Therefore, we propose a rationing policy that uses static thresholds, which can be computed by (12).

The intuition is as follows. By property 1 of Proposition 3, the total number of products reserved for future Class k demand should be no larger than $\eta_{n,t}^*$. It also implies that the total number of products reserved for future Class k demand should be no smaller than $\eta_{k+1,t}^*(\mathbf{0})$. We approximate $\eta_{k+1,t}^*(\mathbf{0})$ using a myopic solution and use this solution as the fixed rationing level in the heuristic. This is a lower bound on the the total number of products reserved in each period under the optimal policy. Thus we call this allocation policy: the minimum fixed rationing policy.

Note that if we reject one unit of demand with lower priority and reserve this unit of product for future demand with higher priority, we incur the cost of $(l_1 + h)$ in the current period and have one more backorder of lower demand class. In the next period, if $D_{t+1}^k \geq \eta_t^H$, this reserved unit will be used to fulfill demand with higher priority and we will have one less backorder of higher demand class and one more backorder of lower demand class. The achieved cost saving is $(l_k - l_1)$. In other cases, this unit will be used to fulfill demand with lower demand class. The fixed rationing level, which is the solution to (12), balances the cost in the current period and potential saving in

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Table 3: Computational performance, CPU time in seconds (s), minutes (m), hours (h), and days (d) ($T = 15$, $\mu^1 = 15$, $\mu^k = 15$, $h = 1$, $l_1 = 3$, $l_k = 7$, $\delta = 5$, and $k = n - 1$)

n		2	3	4	5	6
$L = 0$	Optimal policy	3.1s	84s	29m	9h	>2d
	Ordering policy in the heuristic	0.14s	0.33s	0.49s	0.72s	0.93s
	Allocation policy in the heuristic	<0.1s	<0.1s	<0.1s	<0.1s	<0.1s
$L = 1$	Optimal policy	75s	27m	9h	>2d	>2d
	Ordering policy in the heuristic	2.5s	2.5s	2.7s	2.8s	3.1s
	Allocation policy in the heuristic	<0.1s	<0.1s	<0.1s	<0.1s	<0.1s
$L = 2$	Optimal policy	23m	9h	>2d	>2d	>2d
	Ordering policy in the heuristic	4.3s	4.6s	4.8s	5.5s	6.3s
	Allocation policy in the heuristic	<0.1s	<0.1s	<0.1s	<0.1s	<0.1s

the future.

Ordering policy: Approximated myopic ordering policy, i.e., the order quantity y_t^H in state $(x_{n+L-1}, \ldots, x_1)$ is the optimal solution to the following optimization problem:

$$\begin{aligned}\max_{y\geq 0}\ & l_k E(D_{t+L}^k - \sum_{i=k}^{n} x_{i,L})^+ + l_1 E[D_{t+L}^1 - \sum_{i=1}^{k-1} x_{i,L} - \max\{\sum_{i=k}^{n} x_{i,L} - D_{t+L}^k - \eta_{t+L}^H, (x_{k,L} - D_{t+L}^k)^+\}]^+ \\ & + hE[\sum_{i=1}^{k-1} x_{i,L} + \max\{\sum_{i=k}^{n} x_{i,L} - D_{t+L}^k - \eta_{t+L}^H, (x_{k,L} - D_{t+L}^k)^+\} - D_{t+L}^1]^+ \\ & + hE\min\{\eta_{t+L}^H, (\sum_{i=k}^{n} x_{i,L} - D_{t+L}^k)^+ - (x_{k,L} - D_{t+L}^k)^+\} + \beta^{n-1}\delta o(y),\end{aligned}$$

where $o(y) = E[(\sum_{i=k}^{n} x_{i,L} - \sum_{\tau=t+L}^{t+L+n} D_\tau^k)^+ + (\sum_{i=1}^{k-1} x_{i,L} - \sum_{\tau=t+L}^{t+L+n} D_\tau^1)^+]$ and $x_{i,L}$ is computed by the following recursive equations:

$$\tilde{x}_{i,\tau} = (x_{i,\tau} - (D_\tau^k - \sum_{j=k}^{i-1} x_{j,\tau})^+)^+, \qquad \text{for } i = k, \ldots, n,$$

$$z_{i,\tau} = (\tilde{x}_{i,\tau} - (\eta_{\tau+t}^H - \sum_{j=i+1}^{n+1} \tilde{x}_{j,\tau})^+)^+, \qquad \text{for } i = k+1, \ldots, n,$$

$$z_{k,\tau} = \tilde{x}_{k,\tau},$$

$$x_{1,\tau+1} = (x_{2,\tau} - (D_\tau^1 - x_{1,\tau})^+)^+ - (D_\tau^1 - \sum_{j=1}^{k-1} x_{i,\tau} - \sum_{j=k}^{n} z_{j,\tau})^+,$$

$$x_{k,\tau+1} = \tilde{x}_{k+1,\tau} - z_{k+1,\tau} + (z_{k+1,\tau} - (D_\tau^1 - \sum_{j=1}^{k-1} x_{j,\tau} - z_{k,\tau})^+)^+ - (D_\tau^k - \sum_{j=k}^{n} x_{j,\tau})^+,$$

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$$x_{i,\tau+1} = (x_{i+1,\tau} - (D^1_\tau - \sum_{j=1}^{i} x_{i,\tau})^+)^+, \qquad \text{for } i = 2, \ldots, k-2,$$

$$x_{i,\tau+1} = \tilde{x}_{i+1,\tau} - z_{i+1,\tau} + (z_{i+1,\tau} - (D^1_\tau - \sum_{j=1}^{k-1} x_{j,\tau} - \sum_{j=k}^{i} z_{j,\tau})^+)^+, \qquad \text{for } i = k-1, k+1, \ldots, n-1,$$

$$x_{i,\tau+1} = x_{i+1,\tau}, \qquad \text{for } i = n, \ldots, n+L-1.$$

where $x_{n+L,0} = y$ and $x_{i,0} = x_i$ for $i = 1, \ldots, n+L-1$.

This proposed ordering policy is motivated by Nahmias (1976), which shows that for perishable inventory systems with backorders and a single demand class, the approximated myopic policy performs close to optimal. We consider the approximated myopic solution of our problem. Suppose at the beginning of period t, the net inventory levels are $(x_{n+L-1}, \ldots, x_1)$ and the order quantity is y. Then, the new order y will arrive in period $t+L$ and the net inventory levels will become $(x_{n,L}, \ldots, x_{1,L})$ if the minimum disposal policy and the minimum fixed rationing policy are applied. In this period, following the minimum fixed rationing policy, the number of backorders of Class k demand is $(D^k_{t+L} - \sum_{i=k}^{n} x_{i,L})^+$, the number of products that are reserved into the next period is $\min\{\eta^H_{t+L}, (\sum_{i=k}^{n} x_{i,L} - D^k_{t+L})^+ - (x_{k,L} - D^k_{t+L})^+\}$ (we only reserve products with lifetime larger than k), the number of backorders of Class 1 demand is

$$\begin{aligned}(D^1_{t+L} - \sum_{i=1}^{k-1} x_{i,L} - ((\sum_{i=k}^{n} x_{i,L} - D^k_{t+L})^+ - (x_{k,L} - D^k_{t+L})^+ - \eta^H_{t+L})^+ + (x_{k,L} - D^k_{t+L})^+)^+ \\ = (D^1_{t+L} - \sum_{i=1}^{k-1} x_{i,L} - \max\{\sum_{i=k}^{n} x_{i,L} - D^k_{t+L} - \eta^H_{t+L}, (x_{k,L} - D^k_{t+L})^+\})^+,\end{aligned}$$

the number of products that are carried into the next period is

$$\begin{aligned}(\sum_{i=1}^{k-1} x_{i,L} + \max\{\sum_{i=k}^{n} x_{i,L} - D^k_{t+L} - \eta^H_{t+L}, (x_{k,L} - D^k_{t+L})^+\} - D^1_{t+L})^+ \\ + \min\{\eta^H_{t+L}, (\sum_{i=k}^{n} x_{i,L} - D^k_{t+L})^+ - (x_{k,L} - D^k_{t+L})^+\} - (x_{1,L} - D^1_{t+L})^+,\end{aligned}$$

and the number of products that are disposed is $(x_{1,L} - D^1_{t+L})^+$. Note that $x_{1,L}$ is independent of y. Thus, when minimizing the one-period cost, we can ignore terms that are associated with $(x_{1,L} - D^1_{t+L})^+$. We also need to consider the expected disposal amount for the new order y. Using the same idea in Nahmias (1976), we approximate it by $E[(\sum_{i=k}^{n} x_{i,L} - \sum_{\tau=t+L}^{t+L+n} D^k_\tau)^+ + (\sum_{i=1}^{k-1} x_{i,L} -$

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Table 4: Relative difference between the optimal cost and the cost under the heuristic (%)
($\mu^1 = 10$, $\mu^k = 15$)

			(L,n,k)								
h	l_k	δ	(0,2,1)	(0,3,1)	(0,3,2)	(0,4,1)	(0,4,2)	(1,2,1)	(1,3,1)	(1,3,2)	(2,2,1)
1	4	1	0.65	0.44	0.89	0.34	0.73	1.02	0.68	1.56	1.85
-	-	3	0.72	0.49	1.08	0.37	0.89	1.18	0.81	1.62	2.56
-	-	5	0.81	0.58	1.55	0.45	1.11	1.35	0.97	1.98	2.92
1	7	1	0.80	0.51	1.06	0.42	0.94	1.14	0.88	1.78	2.04
-	-	3	0.88	0.55	1.23	0.45	1.17	1.28	1.00	1.98	2.73
-	-	5	0.98	0.71	1.70	0.55	1.33	1.76	1.20	2.26	3.46
1	10	1	0.94	0.64	1.36	0.51	1.10	1.36	1.11	2.15	2.31
-	-	3	0.98	0.80	1.47	0.63	1.27	1.59	1.22	2.30	2.98
-	-	5	1.05	0.89	1.87	0.72	1.51	2.05	1.58	2.61	3.72
3	4	1	0.39	0.18	0.57	0.08	0.43	0.99	0.57	1.62	1.96
-	-	3	0.46	0.20	0.86	0.12	0.61	1.09	0.85	1.81	2.16
-	-	5	0.53	0.34	1.06	0.21	0.86	1.25	1.03	2.01	2.41
3	7	1	0.48	0.34	0.90	0.19	0.67	1.12	0.87	1.82	2.04
-	-	3	0.52	0.40	1.17	0.20	0.76	1.30	1.00	1.99	2.22
-	-	5	0.67	0.49	1.41	0.27	1.01	1.34	1.09	2.15	2.65
3	10	1	0.70	0.41	1.09	0.18	0.82	1.42	0.95	2.03	2.34
-	-	3	0.72	0.57	1.31	0.25	0.99	1.52	1.16	2.26	2.53
-	-	5	0.86	0.68	1.64	0.40	1.25	1.70	1.44	2.44	3.17
5	4	1	0.15	0.11	0.37	0.10	0.33	0.46	0.37	1.61	1.10
-	-	3	0.21	0.13	0.49	0.10	0.40	0.56	0.50	1.75	1.39
-	-	5	0.26	0.29	0.55	0.13	0.62	0.77	0.61	1.85	1.54
5	7	1	0.24	0.21	0.46	0.16	0.60	0.80	0.67	1.73	1.53
-	-	3	0.29	0.21	0.52	0.17	0.65	0.83	0.75	1.89	1.76
-	-	5	0.37	0.33	0.67	0.18	0.77	0.96	0.85	1.96	1.89
5	10	1	0.39	0.30	0.51	0.21	0.56	0.87	0.70	1.93	2.09
-	-	3	0.44	0.34	0.84	0.24	0.68	0.94	0.82	2.05	2.19
-	-	5	0.54	0.46	1.06	0.32	0.99	1.17	1.07	2.11	2.43

$\sum_{\tau=t+L}^{t+L+n} D_\tau^1)^+]$. Then, the approximated myopic ordering quantity is the optimal solution that minimize the expected one-period cost.

In Table 3, we provide representative results comparing the computational time of the heuristic and the optimal policy. As we can see, the heuristic significantly outperforms the optimal policy in the computational time. This is particularly the case when the leadtime or the lifetime is large.

In Table 4, which includes 243 combinations of parameter, we compare the optimal cost with the cost under the heuristic. As we see, the relative difference between the optimal cost and the cost under the heuristic is under 4% in all the examples. This suggest that the heuristic performs

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consistently well.

6.3 Impacts of Leadtime, Lifetime and Demand Type

In this subsection, we study the second question: how can the firm improve the control of perishable items? To do so, we examine the impacts of the leadtime, the lifetime, and the demand class on the optimal expected total cost. We assume the starting inventory at the beginning of the planning horizon is 0. Given leadtime L, lifetime of the products n, and two demand classes: Class 1 and Class k, we denote the system by "S-(L, n, k)" and the optimal expected total cost by $v_1^{L,n,k}$.

Proposition 6. *$v_1^{L,n,k}$ is nondecreasing in L and k, and is nonincreasing in n.*

Proposition 6 suggests that the firm can reduce the total cost by decreasing the leadtime, increasing the lifetime of the products, or decreasing the acceptance threshold of the demand (increasing customers' willingness to accept older products). This is intuitive since all these strategies can make the products "less perishable". However, a-priori, it is not clear which one of this strategies is more beneficial. We study this below.

Proposition 7. *The benefit of reducing the leadtime is greater than the benefit of increasing the lifetime. The benefit of increasing the lifetime is greater than the benefit of deceasing the acceptance threshold of the demand.*

We numerically examine the magnitude of the benefits of these strategies. In Table 5, we compare the cost reduction under different strategies for different L, n, and k. The following observations can be made. First, the benefit of reducing the leadtime can be significant. Reducing the leadtime by 1 can reduce the cost by 20%. Also, this benefit is more significant when the leadtime is smaller. Second, the benefit of increasing the lifetime of the products is in the middle range, between 4% and 9%. This benefit is smaller when the lifetime is larger. Third, the benefit of decreasing the acceptance threshold of the demand is limited. It is less than 3% for all the cases examined.

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Table 5: Cost reduction under different strategies (%) ($\mu^1 = 10$, $\mu^k = 15$, $h = 1$, $l_1 = 3$, $l_k = 7$, and $\delta = 5$.)

L	n	k	Reducing the leadtime by 1	Increasing the lifetime by 1	Decreasing the acceptance threshold of Class k demand by 1
1	2	2	20.41	9.06	-
1	3	2	17.90	7.39	-
1	3	3	18.06	7.77	1.93
1	4	2	15.61	4.65	-
1	4	3	17.10	5.93	1.72
1	4	4	17.68	6.80	2.65
2	2	2	15.31	8.09	-
2	3	2	14.30	5.62	-
2	3	3	14.54	6.03	2.39
3	2	2	10.67	6.56	-

7 Conclusion

In this paper, we presented a new model for perishable systems with multiple demand classes that require products of different ages. We characterized the structure of the optimal ordering, allocation, and disposal policies. We showed that the optimal order quantity is decreasing in the inventory levels and is more sensitive to the inventory level of fresher products, the optimal allocation policy is a sequential rationing policy, and the optimal disposal policy is characterized by $n-1$ thresholds. We show that the optimal policy is effective and robust. We examined how the firm can improve the control of perishable items and showed that the benefit of decreasing the leadtime is more significant than the benefit of increasing the lifetime of the products or the benefit of decreasing the acceptance threshold of the demand. We also developed an effective heuristic which is at most 4% away from the optimal cost in our numerical examples.

There are several possible avenues for future research. It would be useful to incorporate the pricing decision into our problem. Since there are multiple demand classes with different requirements on the age of acceptable products, it would be of interest to investigate how to set prices for difference demand classes. It would also be interesting to consider a two-echelon perishable inventory system with a warehouse and several distribution centers, where each center requires products of a sufficient remaining shelf life. It would then be useful to examine how to order products for the central warehouse and how to allocation products of different ages to different distribution centers.

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Appendix: Proofs

We employ the concept of multimodularity. The definition of multimodularity and its properties can be found at Altman et al. (2000) and Li and Yu (2014). For the sake of brevity, we omit the proofs of simple lemmas and propositions, including Lemma 1, Propositions 4, 6, and 7. We also omit the proof of Theorem 4, which is similiar to the proofs for the case with two demand classes. These proofs and more details of the proofs below can be found at Abouee-Mehrizi et al. (2017).

We first use induction to prove Lemma 2, Proposition 1, Theorems 1, 2, and 3 together. Specifically, for $t = 1, \ldots, T + L$ we assume that v_{t+1} satisfies properties in Proposition 1; utilize these properties to characterize the optimal ordering, allocation and disposal policies in period t; and then use this structure of the optimal solution to show that these properties hold for v_t. In the following, for $f(x_1, \ldots, x_n)$, we denote $\frac{\partial f}{\partial x_i}(x_1, \ldots, x_n)|_{(x_1,\ldots,x_n)=(y_1,\ldots,y_n)}$ by $\frac{\partial f}{\partial x_i}(y_1, \ldots, y_n)$ for simplicity.

Lemma 3. *If v_{t+1} satisfies properties in Proposition 1, then the optimal disposal policy in period t is described in Theorem 1.*

Proof. When $r_i \geq 0$, $\frac{\partial U_t}{\partial r_i}(x_{n+L}, \ldots, x_{n+1}, \mathbf{r}) = h - \delta + \beta \frac{\partial v_{t+1}}{\partial x_{i-1}}(x_{n+L}, \ldots, x_{n+1}, \mathbf{r})$. Since v_{t+1} satisfies property (2) of Proposition 1, then for $r_i \geq 0$ and $r_j \geq 0$ with $i > j$, we have

$$\frac{\partial U_t}{\partial r_i}(x_{n+L}, \ldots, x_{n+1}, \mathbf{r}) \leq \frac{\partial U_t}{\partial r_j}(x_{n+L}, \ldots, x_{n+1}, \mathbf{r}). \tag{13}$$

Note that we will dispose a unit of product with lifetime i if the derivative of U_t with respect to r_i is positive. This implies that it is optimal to dispose products with lifetime 2 first, then only once there are no products of lifetime 2, disposing product of lifetime 3 should be considered and so on.

Next we show that $r^*_{i,t}$ defined in (7) represents the optimal reservation thresholds for their respective inventory levels. Note that $r^*_{i,t} = \max \operatorname*{argmin}_{r \geq 0} U_t(x_{n+L}, \ldots, x_{n+1}, z_n, \ldots, z_{i+1}, r, \mathbf{0})$. By Proposition 1, $v_{t+1}(x_{n+L-1}, \ldots, x_1)$ is multimodular and thus component-wise convex. Therefore, $U_t(x_{n+L}, \ldots, x_{n+1}, \mathbf{r}) = h \sum_{i=2}^{n} r_i^+ - \delta \sum_{i=2}^{n} r_i + \beta v_{t+1}(x_{n+L}, x_{n+L}, \ldots, x_{n+1}, \mathbf{r})$ is component-wise convex. Moreover, $\lim_{r \to \infty} U_t(x_{n+L}, \ldots, x_{n+1}, z_n, \ldots, z_{i+1}, r, \mathbf{0}) = \infty$. This implies that

$$r^*_{i,t} = \begin{cases} 0, & \text{if } \frac{\partial U_t}{\partial r_i}(x_{n+L}, \ldots, x_{n+1}, \mathbf{r})_{\ \mathbf{r}=(z_n,\ldots,z_{i+1},\mathbf{0})} \geq 0, \\ \hat{r}_{i,t}, & \text{otherwise,} \end{cases}$$

for $i = 2, \ldots, n$, where $\hat{r}_{i,t}$ is the unique solution to $\frac{\partial U_t}{\partial r_i}(x_{n+L}, \ldots, x_{n+1}, \mathbf{r})_{\ \mathbf{r}=(z_n,\ldots,z_{i+1},r,\mathbf{0})} = 0$. To show that $r^*_{i,t}$ defined in (7) represents the optimal reservation thresholds, note that if $z_{k+1,t} < 0$, i.e., there are backorders of Class k demand after demand fulfillment, then $z^+_{i,t} = 0$ for $i = k, \ldots, n$. Therefore, $z_{i,t} \leq r^*_{i,t}$ for $i = k, \ldots, n$ and it is optimal not to dispose any products with lifetime larger than or equal to k. Similarly, if $z_{2,t} < 0$, it is optimal not to dispose any products. Therefore, we only need to consider two cases: (1) $z_{k+1,t} \geq 0$ and $z_{2,t} \geq 0$, and (2) $z_{k+1,t} < 0$ and $z_{2,t} \geq 0$.

For the first case, let $i^* = \max\{i : z_i \geq r^*_{i,t}\}$. Then, we have $z_i < r^*_{j,t}$ for $j > i^*$. Since v_{t+1} satisfies property (1) of Proposition 1, it is multimodular and thus jointly convex and supermodular. Therefore, $U_t(x_{n+L}, \ldots, x_{n+1}, \mathbf{r}) = h \sum_{i=2}^{n} r_i^+ - \delta \sum_{i=2}^{n} r_i + \beta v_{t+1}(x_{n+L}, x_{n+L}, \ldots, x_{n+1}, \mathbf{r})$ is jointly convex and supermodular. This implies that $\frac{\partial U_t}{\partial r_i}(x_{n+L}, \ldots, x_{n+1}, \mathbf{r})$ is increasing in r_j, for $i, j = 2, \ldots, n$. As a consequence, for any $r_{i^*} \geq r^*_{i^*,t}$, $r_j \geq 0$, $j = 1, \ldots, i^* - 1$, and $i \leq i^*$, we have

$$\begin{aligned} \frac{\partial U_t}{\partial r_i}(x_{n+L}, \ldots, x_{n+1}, z_n, \ldots, z_{i^*+1}, r_{i^*}, \ldots, r_2) &\geq \frac{\partial U_t}{\partial r_{i^*}}(x_{n+L}, \ldots, x_{n+1}, z_n, \ldots, z_{i^*+1}, r_{i^*}, \ldots, r_2) \\ &\geq \frac{\partial U_t}{\partial r_{i^*}}(x_{n+L}, \ldots, x_{n+1}, z_n, \ldots, z_{i^*+1}, r^*_{i^*,t}, \mathbf{0}) \geq 0, \end{aligned}$$

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where the first inequality is by (13), the second inequality is due to the fact that $\frac{\partial U_t}{\partial r_i}(x_{n+L},\dots,x_{n+1},\mathbf{r})$ is increasing in r_j for all i,j, and the last one is by the definition of $r^*_{i,t}$. Similarly, for any $\tilde{\mathbf{r}} < (z_n,\dots,z_{i^*+1},r^*_{i^*,t},0,\dots,0)$, if $r^*_{i^*,t}>0$, then for any $i \geq i^*$, we have

$$\frac{\partial U_t}{\partial r_i}(x_{n+L},\dots,x_{n+1},\tilde{\mathbf{r}}) \leq \frac{\partial U_t}{\partial r_{i^*}}(x_{n+L},\dots,x_{n+1},\tilde{\mathbf{r}}) \leq \frac{\partial U_t}{\partial r_{i^*}}(x_{n+L},\dots,x_{n+1},z_n,\dots,z_{i^*+1},r^*_{i^*,t},\mathbf{0}) = 0.$$

If $r^*_{i^*,t}=0$, by the definition of i^*, we have $r_{i^*+1,t} > z_{i^*+1} \geq 0$. Thus, for any $i \geq i^*+1$,

$$\begin{aligned}\frac{\partial U_t}{\partial r_i}(x_{n+L},\dots,x_{n+1},\tilde{\mathbf{r}}) &\leq \frac{\partial U_t}{\partial r_{i^*+1}}(x_{n+L},\dots,x_{n+1},\tilde{\mathbf{r}})\\ &\leq \frac{\partial U_t}{\partial r_{i^*+1}}(x_{n+L},\dots,x_{n+1},z_n,\dots,z_{i^*+2},r_{i^*+1,t},\mathbf{0}) = 0.\end{aligned}$$

This implies that it is optimal to dispose all the products with lifetime less than i^* and keep $r^*_{i^*,t}$ units of products with lifetime i^* and all younger products (i.e. those with lifetime $i^*+1,\dots,n$). In other words, it is optimal to keep $\min\{z_{i,t},r^*_{i,t}\}$ units of products with lifetime i.

For the second case, it is optimal not to dispose any products with lifetime larger than or equal to k. Similar to the proof of case (1), we can show that it is optimal to keep $\min\{z_{i,t},r^*_{i,t}\}$ units of products with lifetime i for $i=2,\dots,k-1$.

Last, we show that $r^*_{i,t}$ satisfies properties (1) and (2) in Theorem 1. Property (1) directly follows from the definition of $r^*_{i,t}$. If $z_i > r^*_{i,t}$, then for any $j<i$, we have

$$\frac{\partial U_t}{\partial r_j}(x_{n+L},\dots,x_{n+1},z_n,\dots,z_{j+1},0,\dots,0) \geq \frac{\partial U_t}{\partial r_i}(x_{n+L},\dots,x_{n+1},z_n,\dots,z_{i+1},r^*_{i,t},0,\dots,0) \geq 0.$$

By the definition of $r^*_{j,t}$, we have $r^*_{j,t}=0$. Therefore, $r^*_{i,t}$ satisfies property (2). □

Lemma 4. *If v_{t+1} satisfies properties in Proposition 1, then $u_t(x_{n+L},x_{n+L-1},\dots,x_{n+1},\mathbf{z})$ satisfies the following properites:*
(1) $u_t(x_{n+L},x_{n+L-1},\dots,x_{n+1},\mathbf{z})$ is jointly convex;
(2) $\frac{\partial u_t}{\partial z_i} \leq \frac{\partial u_t}{\partial z_j}$ for $i>j$;
(3) $\frac{\partial u_t}{\partial z_i} \leq \delta$ when $z_i \geq 0$;
(4) $\frac{\partial u_t}{\partial z_k} \geq \frac{\partial u_t}{\partial z_2} - l_1$ and $\frac{\partial u_t}{\partial z_n} \geq \frac{\partial u_t}{\partial z_{k+1}} - l_k$.

Proof. (1) By property (1) of Proposition 1, v_{t+1} is multimodular and thus jointly convex. Therefore, $U_t(x_{n+L},\dots,x_{n+1},\mathbf{r}) = h\sum_{i=2}^n r_i^+ - \delta\sum_{i=2}^n r_i + \beta v_{t+1}(x_{n+L},x_{n+L},\dots,x_{n+1},\mathbf{r})$ is jointly convex. To show $u_t(x_{n+L},x_{n+L-1},\dots,x_{n+1},\mathbf{z})$ is jointly convex, we need to show that for $\lambda\in[0,1]$,

$$\lambda u_t(x_{n+L},\dots,x_{n+1},\mathbf{z}) + (1-\lambda)u_t(\bar{x}_{n+L},\dots,\bar{x}_{n+1},\bar{\mathbf{z}}) \geq u_t(\hat{x}_{n+L},\dots,\hat{x}_{n+1},\hat{\mathbf{z}}), \tag{14}$$

where $\hat{x}_i = \lambda x_i + (1-\lambda)\bar{x}_i$ for $i=n+1,\dots,n+L$ and $\hat{\mathbf{z}} = \lambda\mathbf{z}+(1-\lambda)\bar{\mathbf{z}}$. Let $r^* = \min_{\min\{\mathbf{0},\mathbf{z}\}\leq\mathbf{r}\leq\mathbf{z}} U_t(x_{n+L},\dots,x_{n+1},\mathbf{r})$ and $\bar{r}^* = \min_{\min\{\mathbf{0},\bar{\mathbf{z}}\}\leq\mathbf{r}\leq\bar{\mathbf{z}}} U_t(\bar{x}_{n+L},\dots,\bar{x}_{n+1},\bar{\mathbf{r}})$. We consider three cases: (a) $z_{k+1}\geq 0$ and $\bar{z}_{k+1}\geq 0$, (b) $z_{k+1}\geq 0$ and $\bar{z}_{k+1}<0$, and (c) $z_{k+1}<0$ and $\bar{z}_{k+1}<0$.

For the first case, we again consider three cases: (i) $z_2\geq 0$ and $\bar{z}_2\geq 0$, (ii) $z_2<0$ and $\bar{z}_2<0$, and (iii) $z_2\geq 0$ and $\bar{z}_2<0$. In case (i), we have $0\leq r^*_i\leq z_i$ and $0\leq\bar{r}^*_i\leq\bar{z}_i$ for $i=2,\dots,n$. Thus, $\lambda\mathbf{r}^*+(1-\lambda)\bar{\mathbf{r}}^*$ is a feasible solution for the state $(\hat{x}_{n+L},\dots,\hat{x}_{n+1},\hat{\mathbf{z}})$. Therefore,

$$\begin{aligned}&\lambda u_t(x_{n+L},\dots,x_{n+1},\mathbf{z}) + (1-\lambda)u_t(\bar{x}_{n+L},\dots,\bar{x}_{n+1},\bar{\mathbf{z}})\\ &\quad= \lambda(\delta\sum_{i=2}^n z_i + U_t(x_{n+L},\dots,x_{n+1},\mathbf{r}^*)) + (1-\lambda)(\delta\sum_{i=2}^n\bar{z}_i + U_t(\bar{x}_{n+L},\dots,\bar{x}_{n+1},\bar{\mathbf{r}}^*))\\ &\quad\geq \delta\sum_{i=2}^n(\lambda z_i+(1-\lambda)\bar{z}_i) + U_t((\hat{x}_{n+L},\dots,\hat{x}_{n+1},\lambda\mathbf{r}^*+(1-\lambda)\bar{\mathbf{r}}^*)\\ &\quad\geq u_t((\hat{x}_{n+L},\dots,\hat{x}_{n+1},\lambda\mathbf{z}+(1-\lambda)\bar{\mathbf{z}}),\end{aligned}$$

where the first inequality is due to the fact that U_t is jointly convex and the second inequality is due to the fact that $\lambda\mathbf{r}^*+(1-\lambda)\bar{\mathbf{r}}^*$ is a feasible solution. We can similarly show (14) holds in cases

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(ii) and (iii). The proofs of cases (b) and (c) are also similar.

(2) It is equivalent to show that for $i > j$ and for small enough ϵ,

$u_t(x_{n+L}, \ldots, x_n, z_n, \ldots, z_{i+1}, z_i+\epsilon, z_{i-1}, \ldots, z_2) \leq u_t(x_{n+L}, \ldots, x_n, z_n, \ldots, z_{j+1}, z_j+\epsilon, z_{j-1}, \ldots, z_2)$.

Let $\bar{\mathbf{z}} = (z_n, \ldots, z_{j+1}, z_j + \epsilon, z_{j-1}, \ldots, z_2)$, $\hat{\mathbf{z}} = (z_n, \ldots, z_{i+1}, z_i + \epsilon, z_{i-1}, \ldots, z_2)$, and $\bar{\mathbf{r}} = \underset{\min\{\mathbf{0},\bar{\mathbf{z}}\}\leq\mathbf{r}\leq\bar{\mathbf{z}}}{\operatorname{argmin}} U_t(x_{n+L}, \ldots, x_{n+1}, \mathbf{r})$. We consider 4 cases: (a) $z_i \geq 0$ and $z_j \geq 0$, (b) $z_i < 0$ and $z_j \geq 0$, (c) $z_i \geq 0$ and $z_j < 0$, and (d) $z_i < 0$ and $z_j < 0$.

(a) If $\bar{r}_j < \epsilon$, let $\hat{r}_j = 0$, $\hat{r}_i = \bar{r}_i + \bar{r}_j$ and $\hat{r}_m = \bar{r}_m$ for $m \neq i$ and $m \neq j$. Then, $\min\{\mathbf{0}, \hat{\mathbf{z}}\} \leq \hat{\mathbf{r}} \leq \hat{\mathbf{z}}$. Note that $\bar{\mathbf{r}} = (\bar{r}_n, \ldots \bar{r}_{j+1}, 0, \bar{r}_{j-1}, \ldots, \bar{r}_2) + \bar{r}_j \mathbf{e}_j$ and $\hat{\mathbf{r}} = (\bar{r}_n, \ldots \bar{r}_{j+1}, 0, \bar{r}_{j-1}, \ldots, \bar{r}_2) + \bar{r}_j \mathbf{e}_i$. As a consequence,

$$\begin{aligned} u_t(x_{n+L}, \ldots, x_n, \bar{\mathbf{z}}) - u_t(x_{n+L}, \ldots, x_n, \hat{\mathbf{z}}) &\geq U_t(x_{n+L}, \ldots, x_{n+1}, \bar{\mathbf{r}}) - U_t(x_{n+L}, \ldots, x_{n+1}, \hat{\mathbf{r}}) \\ &= \beta v_{t+1}(x_{n+L}, \ldots, x_{n+1}, \bar{\mathbf{r}}) - \beta v_{t+1}(x_{n+L}, \ldots, x_{n+1}, \hat{\mathbf{r}}) \geq 0, \end{aligned}$$

where the first inequality is because $u_t(x_{n+L}, \ldots, x_{n+1}, \bar{\mathbf{z}}) = \delta \sum_{i=2}^{n} \bar{z}_i + U_t(x_{n+L}, \ldots, x_{n+1}, \bar{\mathbf{r}})$, $u_t(x_{n+L}, \ldots, x_{n+1}, \hat{\mathbf{z}}) \leq \delta \sum_{i=2}^{n} \hat{z}_i + U_t(x_{n+L}, \ldots, x_{n+1}, \hat{\mathbf{r}})$, and $\sum_{i=2}^{n} \bar{z}_i = \sum_{i=2}^{n} \hat{z}_i$, and the second inequality is because $\frac{\partial v_{t+1}}{\partial x_{i-1}} \leq \frac{\partial v_{t+1}}{\partial x_{j-1}}$ by property (2) of Proposition 1.

If $\bar{r}_j \geq \epsilon$, let $\hat{r}_j = \bar{r}_j - \epsilon$, $\hat{r}_i = \bar{r}_i + \epsilon$ and $\hat{r}_m = \bar{r}_m$ for $m \neq i$ and $m \neq j$. Then, $\min\{\mathbf{0}, \hat{\mathbf{z}}\} \leq \hat{\mathbf{r}} \leq \hat{\mathbf{z}}$. As a consequence,

$u_t(x_{n+L}, \ldots, x_{n+1}, \bar{\mathbf{z}}) - u_t(x_{n+L}, \ldots, x_{n+1}, \hat{\mathbf{z}}) \geq U_t(x_{n+L}, \ldots, x_{n+1}, \bar{\mathbf{r}}) - U_t(x_{n+L}, \ldots, x_{n+1}, \hat{\mathbf{r}}) \geq 0.$

Therefore, we have $\frac{\partial u_t}{\partial z_i} \leq \frac{\partial u_t}{\partial z_j} \leq \delta$ when $z_i \geq 0$ and $z_j \geq 0$ for $i > j$;

The proofs of (b), (c), and (d) are similar.

(3) The proofs of parts (3) and (4) are similar to the proof of part (2). □

Lemma 5. *If v_{t+1} satisfies properties in Proposition 1, then the optimal allocation policy in period t is described in Theorem 2.*

Proof. Note that we will use one unit of product with lifetime i to satisfy one unit of Class j demand if the derivative of W_t with respect to a_i^j is negative. For $j \in \{1, k\}$ and $i = 2, \ldots, n$, we have $\frac{\partial W_t}{\partial a_1^j} = -\delta - l_j + \frac{\partial u_t}{\partial z_{j+1}}$, and $\frac{\partial W_t}{\partial a_i^j} = -l_j - \frac{\partial u_t}{\partial z_i} + \frac{\partial u_t}{\partial z_{j+1}}$. By properties (2) and (4) of Lemma 4, for $i = 1, \ldots, n$, $\frac{\partial W_t}{\partial a_i^1} > \frac{\partial W_t}{\partial a_i^k}$. This implies that it is optimal to fulfill Class k demand first. By property (2) of Lemma 4, for $i < j$, $\frac{\partial W_t}{\partial a_i^k} \leq \frac{\partial W_t}{\partial a_j^k} \leq 0$. This implies that it is optimal to use products with lifetime i to fulfill Class k demand before using products with lifetime j for $i < j$. In other words, it is optimal to first satisfy Class k demand as much as possible following the FIFO policy (using products with lifetime longer than k).

Given the posterior net inventory levels $(\tilde{x}_{n+L}, \ldots, \tilde{x}_{n+1}, \tilde{x}_n, \ldots, \tilde{x}_1)$ after fulfilling Class k demand, to choose products of which lifetime to satisfy Class 1 demand, we need to solve the following optimization problem

$$\begin{aligned} \min\ & \delta(x_1 - a_1^1) + l_1(d^1 + \tilde{x}_1^- - \sum_{i=1}^{n} a_i^1) + u_t(\tilde{x}_{n+L}, \ldots, \tilde{x}_{n+1}, \tilde{x}_n - a_n^1, \ldots, \tilde{x}_2 - a_2^1 - (d^1 + \tilde{x}_1^- - \sum_{i=1}^{n} a_i^1)) \\ s.t.\ & 0 \leq a_i^1 \leq \tilde{x}_i^+, \quad i = 1, \ldots, n \\ & \sum_{i=1}^{n} a_i^1 \leq \tilde{x}_1^- + d^1, \end{aligned}$$

We denote the objective function by $\tilde{W}_t(\tilde{x}_{n+L}, \ldots, \tilde{x}_1, d^1, \mathbf{a}^1)$. By Lemma 4, we have $\frac{\partial \tilde{W}_t}{\partial a_i^1} \leq \frac{\partial \tilde{W}_t}{\partial a_j^1}$

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for $i < j$ and $\frac{\partial \tilde{W}_t}{\partial a_i^1} \leq 0$ when $i \leq k$. The first inequality implies that it is optimal to fulfill Class 1 demand following the FIFO policy and the second inequality implies that it is optimal to use products with lifetime smaller than or equal to k to satisfy Class 1 demand as much as possible.

Similar to the proof of Lemma 3, we can show that $\eta_{i,t}^*$ defined in (8) represent the optimal rationing thresholds for their respective inventory levels when fulfilling Class 1 demand and it satisfies properties (1) and (2) in Theorem 2. □

Lemma 6. *If v_{t+1} satisfies properties in Proposition 1, then $w_t(x_{n+L}, x_{n+L-1}, \ldots, x_1, d^k, d^1)$ satisfies the following properties:*
(1) w_t is multimodular in $(x_{n+L}, \ldots, x_1)$;
(2) $\frac{\partial w_t}{\partial x_n} \leq \frac{\partial w_t}{\partial x_{n-1}} \leq \ldots, \leq \frac{\partial w_t}{\partial x_1} \leq \delta$;
(3) $\frac{\partial w_t}{\partial x_n} \geq \frac{\partial w_t}{\partial x_k} - l_k$ and $\frac{\partial w_t}{\partial x_{k-1}} \geq \frac{\partial w_t}{\partial x_1} - l_1$.

Proof. (1) We first show that optimality equations (5) and (6) are equivalent to two relaxed problems respectively. Then, we combine the two relaxed equations and use the combined equation to show that w_t is multimodular in $(x_{n+L}, \ldots, x_1)$.

We first show that

$$w_t(x_{n+L}, x_{n+L-1}, \ldots, x_1, d^k, d^1) = \min_{\substack{\mathbf{a}^k + \mathbf{a}^1 \leq \mathbf{x}^+ \\ \sum_{i=1}^n a_i^j \leq d^j + x_j^-, j \in \{1,k\} \\ a_i^k = 0, i = 1, \ldots, k-1 \\ \sum_{m=i}^n a_m^j \geq 0, i = j+1, \ldots, n, j \in \{1,k\}}} W_t(x_{n+L}, \ldots, x_1, d^k, d^1, \mathbf{a}^k, \mathbf{a}^1). \quad (15)$$

Since $\frac{\partial W_t}{\partial a_k^k} \leq \frac{\partial W_t}{\partial a_{k+1}^k} \leq \cdots \leq \frac{\partial W_t}{\partial a_n^k} \leq 0$ and $a_i^k = 0$ for $i = 1, \ldots, k-1$, we can first remove the constraint $a_k^k \geq 0$ in (5) and then change the constraints $a_i^k \geq 0$ for $i = k+1, \ldots, n$ in (5) with $\sum_{m=i}^n a_m^k \geq 0$ for $i = k+1, \ldots, n$. We can also remove the constraint $a_1^1 \geq 0$ in (5) since $\frac{\partial W_t}{\partial a_1^1} \leq \frac{\partial W_t}{\partial a_i^1}$ for $i = 2, \ldots, n$ and $\frac{\partial W_t}{\partial a_1^1} \leq 0$. Therefore, to show (15), we only need to show

$$\min_{\substack{\mathbf{a}^k + \mathbf{a}^1 \leq \mathbf{x}^+ \\ \sum_{i=1}^n a_i^j \leq d^j + x_j^-, j \in \{1,k\} \\ a_i^k = 0, i = 1, \ldots, k-1 \\ \sum_{m=i}^n a_m^j \geq 0, i = j+1, \ldots, n, j \in \{1,k\}}} W_t(x_{n+L}, \ldots, x_1, d^k, d^1, \mathbf{a}^k, \mathbf{a}^1) = \min_{\substack{\mathbf{a}^k + \mathbf{a}^1 \leq \mathbf{x}^+ \\ \sum_{i=1}^n a_i^j \leq d^j + x_j^-, j \in \{1,k\} \\ a_i^k = 0, i = 1, \ldots, k-1 \\ \sum_{m=i}^n a_m^k \geq 0, i = k+1, \ldots, n \\ a_i^1 \geq 0, i = 2, \ldots, n}} W_t(x_{n+L}, \ldots, x_1, d^k, d^1, \mathbf{a}^k, \mathbf{a}^1). \quad (16)$$

We denote the feasible region of the LHS of (16) by $\mathbf{S}^L$ and the feasible region of the RHS of (16) by $\mathbf{S}^R$. Since $\mathbf{S}^R \subseteq \mathbf{S}^L$, if we can show that for any $(\mathbf{a}^k, \mathbf{a}^1) \in \mathbf{S}^L$, there exits an $(\tilde{\mathbf{a}}^k, \tilde{\mathbf{a}}^1) \in \mathbf{S}^R$ such that $W_t(x_{n+L}, \ldots, x_1, d^k, d^1, \mathbf{a}^k, \mathbf{a}^1) \geq W_t(x_{n+L}, \ldots, x_1, d^k, d^1, \tilde{\mathbf{a}}^k, \tilde{\mathbf{a}}^1)$, then (16) holds and thus (15) holds. Clearly, if $a_i^1 \geq 0$ for $i = 2, \ldots, n$, we can choose $(\tilde{\mathbf{a}}^k, \tilde{\mathbf{a}}^1) = (\mathbf{a}^k, \mathbf{a}^1)$. Otherwise, let $\hat{i} = \max\{i : a_i^1 < 0\}$. Then, $\hat{i} < n$ and $a_i^1 \geq 0$ for $i = \hat{i}+1, \ldots, n$. Note that $\sum_{i=\hat{i}}^n a_i^1 \geq 0$. We can find $(\epsilon_{\hat{i}+1}, \ldots, \epsilon_n)$ such that $\sum_{i=\hat{i}+1}^n \epsilon_i = -a_{\hat{i}}^1$ and $0 \leq \epsilon_i \leq a_i^1$ for $i = \hat{i}+1, \ldots, n$. In the following, we consider two cases: (1) $\hat{i} < k$, and (2) $\hat{i} \geq k$.

For the first case, we let $\hat{a}_i^1 = a_i^1$ for $i = 1, \ldots, \hat{i}-1$, $\hat{a}_{\hat{i}}^1 = 0$, and $\hat{a}_i^1 = a_i^1 - \epsilon_i$ for $i = \hat{i}+1, \ldots, n$. Furthermore, we let $\hat{a}_i^k = a_i^k$ for $i = 1, \ldots, n$. Then, we have $\hat{a}_i^k + \hat{a}_i^1 = a_i^k + a_i^1 \leq x_i$ for $i = 1, \ldots, \hat{i}-1$,

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$\hat{a}_{\hat{i}}^k+\hat{a}_{\hat{i}}^1=0\le x_{\hat{i}}$, and $\hat{a}_i^k+\hat{a}_i^1\le a_i^k+a_i^1\le x_i$ for $i=\hat{i}+1,\ldots,n$. We also have $\sum_{i=1}^{n}\hat{a}_i^j=\sum_{i=1}^{n}a_i^j\le d^j+x_j^-$ for $j=1,k$, $\hat{a}_i^k=a_i^k=0$ for $i=1,\ldots,k-1$, and $\sum_{m=i}^{n}\hat{a}_m^j\ge 0$ for $i=1,\ldots,n$ and $j=1,k$. Thus, $(\hat{\mathbf{a}}^k,\hat{\mathbf{a}}^1)\in\mathbf{S}^L$. Moreover,

$$\begin{aligned}
&W_t(x_{n+L},\ldots,x_1,d^k,d^1,\hat{\mathbf{a}}^k,\hat{\mathbf{a}}^1)\\
&\quad=W_t(x_{n+L},\ldots,x_1,d^k,d^1,\hat{\mathbf{a}}^k,a_n^1-\epsilon_n,\ldots,a_{\hat{i}+1}^1-\epsilon_{\hat{i}+1},a_{\hat{i}}^1+\sum_{i=\hat{i}}^{n}\epsilon_i,a_{\hat{i}-1}^1,\ldots,a_1)\\
&\quad\le W_t(x_{n+L},\ldots,x_1,d^k,d^1,\hat{\mathbf{a}}^k,a_n^1,\ldots,a_{\hat{i}+1}^1,a_{\hat{i}}^1,a_{\hat{i}-1}^1,\ldots,a_1)\\
&\quad=W_t(x_{n+L},\ldots,x_1,d^k,d^1,\mathbf{a}^k,\mathbf{a}^1),
\end{aligned}$$

where the inequality is due to the fact that $\frac{\partial W_t}{\partial a_1^1}\le\frac{\partial W_t}{\partial a_2^1}\le\cdots\le\frac{\partial W_t}{\partial a_n^1}$.

Using the above approach, we can change the negative components of $\mathbf{a}^1$ one by one from the largest index. That is, we can find an $(\tilde{\mathbf{a}}^k,\tilde{\mathbf{a}}^1)\in\mathbf{S}^L$ with $\tilde{a}_i^1\ge 0$ for $i=1,\ldots,n$, or equivalently an $(\tilde{\mathbf{a}}^k,\tilde{\mathbf{a}}^1)\in\mathbf{S}^R$, such that $W_t(x_{n+L},\ldots,x_1,\mathbf{a}^k,\mathbf{a}^1)\ge W_t(x_{n+L},\ldots,x_1,\tilde{\mathbf{a}}^k,\tilde{\mathbf{a}}^1)$.

For the second case, the proof is similar. This completes the proof for (15).

Using the same approach, we can show that

$$\begin{aligned}
u_t(x_{n+L},x_{n+L-1},\ldots,x_{n+1},\mathbf{z})&=\min_{\mathbf{0}\le\mathbf{r}\le\mathbf{z}^+}\{h\sum_{i=2}^{n}r_i+\delta\sum_{i=2}^{n}(z_i^+-r_i)+\beta v_{t+1}(x_{n+L},\ldots,x_{n+1},\mathbf{r}-\mathbf{z}^-)\}\\
&=\min_{\substack{\sum_{m=i}^{n}r_m\le\sum_{m=i}^{n}z_m^+,i=2,\ldots,n\\ r_i\ge 0,i=2,\ldots,n}}\{h\sum_{i=2}^{n}r_i+\delta\sum_{i=2}^{n}(z_i^+-r_i)+\beta v_{t+1}(x_{n+L},\ldots,x_{n+1},\mathbf{r}-\mathbf{z}^-)\}. \qquad (17)
\end{aligned}$$

Combining (15) and (17), we have

$$\begin{aligned}
w_t(x_{n+L},\ldots,x_1,d^k,d^1)=\min_{\substack{\mathbf{a}^k+\mathbf{a}^1\le\mathbf{x}^+\\ \sum_{i=1}^{n}a_i^j\le d^j+x_j^-,j\in\{1,k\}\\ a_i^k=0,i=1,\ldots,k-1\\ \sum_{m=i}^{n}a_m^j\ge 0,i=j+1,\ldots,n,j\in\{1,k\}\\ \sum_{m=i}^{n}r_m\le\sum_{m=i}^{n}(x_m^+-\sum_{j\in\{1,k\}}a_m^j),i=2,\ldots,n\\ r_i\ge 0,i=2,\ldots,n}}&\{\delta(x_1^+-\sum_{j\in\{1,k\}}a_1^j+\sum_{i=2}^{n}(x_i^+-\sum_{j\in\{1,k\}}a_i^j-r_i))\\
&+h\sum_{i=2}^{n}r_i+\sum_{j\in\{1,k\}}l_j\hat{e}^j+\beta v_{t+1}(x_{n+L},\ldots,x_{n+1},r_n,\ldots,r_{k+1}-\hat{e}^k,r_k,\ldots,r_2-\hat{e}^1)\ \}, \qquad (18)
\end{aligned}$$

where $\hat{e}^j=d^j+x_j^--\sum_{i=1}^{n}a_i^j$ for $j=1,k$. By the structure of the optimal allocation and disposal policy, the optimal solution should also satisfies the following two conditions: (1) if $d^k+x_k^--\sum_{i=1}^{n}a_i^k>0$, then $a_i^k+a_i^1=x_i^+$ for $i=k,\ldots,n$ and $r_i=0$ for $i=k+1,\ldots,n$, and (2) if $d^1+x_1^--\sum_{i=1}^{n}a_i^1>0$, then $a_i^k+a_i^1=x_i^+$ for $i=1,\ldots,k$, $r_i=0$ for $i=2,\ldots,k$, and $\sum_{m=k+1}^{n}r_m=\sum_{m=k+1}^{n}(x_m^+-a_m^k-a_m^1)$. The first optimality condition says that after demand fulfillment, if there are backorders of Class k demand, then the inventory level of products of lifetime i is 0 for $i=k+1,\ldots,n$ and it is optimal not to dispose any products with lifetime larger than k . The second optimality condition says that after demand fulfillment, if there are backorders of Class 1 demand, then the inventory level of products of lifetime i is 0 for $i=1,\ldots,k$ and it is optimal not to dispose any products.

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Next we use (18) to show that w_t is multimodular in $(x_{n+L},\dots,x_1)$. To show this, we show that $\tilde{w}_t(x_{n+L},\dots,x_1,y,d^k,d^1) = w_t(x_{n+L}-y, x_{n+L-1}-x_{n-L},\dots,x_1-x_2,d^k,d^1)$ is submodular in $(x_{n+L},\dots,x_1,y)$. By (18), we have

$$\begin{aligned}\tilde{w}_t(x_{n+L},\dots,x_1,y,d^k,d^1) = \min_{\substack{a_i^k+a_i^1\le(x_i-x_{i+1})^+,i=1,\dots,n\\ \sum_{i=1}^n a_i^j\le d^j+(x_j-x_{j+1})^-,j\in\{1,k\}\\ a_i^k=0,i=1,\dots,k-1\\ \sum_{m=i}^n a_m^j\ge 0,i=j+1,\dots,n,j\in\{1,k\}\\ \sum_{m=i}^n r_m\le\sum_{m=i}^n((x_m-x_{m+1})^+-\sum_{j\in\{1,k\}}a_m^j),i=2,\dots,n\\ r_i\ge 0,i=2,\dots,n}} &\Big\{\delta\Big(\sum_{i=2}^n((x_i-x_{i+1})^+-\sum_{j\in\{1,k\}}a_i^j-r_i)\Big)\\ &+\delta((x_1-x_2)^+-\sum_{j\in\{1,k\}}a_1^j)+h\sum_{i=2}^n r_i+\sum_{j\in\{1,k\}}l_j\bar{e}^j\\ &+\beta v_{t+1}(x_{n+L}-y,\dots,x_{n+1}-x_{n+2},r_n,\dots,r_{k+1}-\bar{e}^k,r_k,\dots,r_2-\bar{e}^1)\Big\}.\end{aligned}$$

where $\bar{e}^j = d^j + (x_j - x_{j+1})^- - \sum_{i=1}^n a_i^j$ for $j = 1, k$. Note that $(x_i - x_{i+1})^+ = x_i - x_{i+1}$ for $i = 2,\dots,k-1,k+1,\dots,n$ and $a_i^k = 0$, for $i = 1,\dots,k-1$. Thus, the constraints become

$$a_i^k + a_i^1 \le x_i - x_{i+1}, \quad i = k+1,\dots,n, \quad (C1) \qquad a_k^k + a_k^1 \le (x_k - x_{k+1})^+, \quad (C2)$$

$$a_i^1 \le x_i - x_{i+1}, \quad i = 2,\dots,k-1, \quad (C3) \qquad a_1^1 \le (x_1 - x_2)^+, \quad (C4)$$

$$\sum_{i=j}^n a_i^j \le d^j + (x_j - x_{j+1})^-, \quad j \in \{1,k\}, \quad (C5) \qquad \sum_{m=i}^n a_m^1 \ge 0, \quad i = 2,\dots,n, \quad (C6)$$

$$\sum_{m=i}^n a_m^j \ge 0, \quad i = k+1,\dots,n, \quad (C7)$$

$$\sum_{m=i}^n r_m \le x_i - x_{n+1} - \sum_{m=i}^n \sum_{j\in\{1,k\}} a_m^j, \quad i = k+1,\dots,n, \quad (C8)$$

$$\sum_{m=i}^n r_m \le x_i - x_{n+1} + (x_k - x_{k+1})^- - \sum_{m=k}^n a_m^k - \sum_{m=i}^n a_m^1, \quad i = 2,\dots,k, \quad (C9)$$

$$r_i \ge 0, \quad i = 2,\dots,n, \quad (C10)$$

and the optimality conditions can be rewritten as (1) if $\bar{e}^k > 0$, then $a_i^k + a_i^1 = x_i - x_{i+1}$ for $i = k+1,\dots,n$ and $r_i = 0$ for $i = k+1,\dots,n$, and (2) if $\bar{e}^1 > 0$, $a_i^1 = x_i - x_{i+1}$ for $i = 2,\dots,k-1$, $a_k^k + a_k^1 = (x_k - x_{k+1})^+$, $r_m = 0$ for $i = 2,\dots,k$, and $\sum_{m=k+1}^n r_m = x_{k+1} - x_{n+1} - \sum_{m=k+1}^n \sum_{j\in\{1,k\}} a_m^j$.

Let $s_i^k = x_i - \sum_{m=i}^n \sum_{j\in\{1,k\}} a_m^j$ for $i = k+1,\dots,n$, $s_k^k = s_{k+1}^k - \bar{e}^k = x_{k+1} - (x_k - x_{k-1})^- + a_k^k - \sum_{m=k+1}^n a_m^1 - d^k$, $s_i^1 = x_i - \sum_{m=i}^n a_m^1$ for $i = 2,\dots,n$, and $s_1^1 = s_2^1 - \bar{e}^1 = x_2 - (x_1 - x_2)^- + a_1^1 - d^1$. Let $\tilde{r}_i = x_{n+1} + \sum_{m=i}^n r_m$ for $i = k+2,\dots,n$, $\tilde{r}_i = x_{n+1} + \sum_{m=i}^n r_m - \bar{e}^k$ for $i = 3,\dots,k+1$, and $\tilde{r}_2 = x_{n+1} + \sum_{m=2}^n r_m - \bar{e}^k - \bar{e}^1$. It can be verified that there is a one-to-one correspondence between $(a_n^k,\dots a_k^k, a_n^1,\dots,a_1^1, r_n,\dots,r_2)$ and $(s_n^k,\dots s_k^k, s_n^1,\dots,s_1^1,\tilde{r}_n,\dots,\tilde{r}_2)$. Then, the constraints become

$$s_n^k \ge x_{n+1}, \quad (C1: i = n) \qquad s_i^k \ge s_{i+1}^k, \quad i = k+1,\dots,n-1, \quad (C1: i \ne n)$$

$$s_k^k \le s_k^1 - d^k, \quad (C2) \qquad s_i^1 \ge s_{i+1}^1, \quad i = 2,\dots,k-1, \quad (C3)$$

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$$s_1^1 \le x_1 - d^1, \qquad (C4) \qquad s_j^j \le s_{j+1}^j, \quad j \in \{1,k\} \qquad (C5)$$
$$s_i^1 \le x_i, \quad i = 2,\dots,n, \qquad (C6) \qquad s_i^k \le s_i^1, \quad i = k+1,\dots,n, \qquad (C7)$$
$$\tilde{r}_i \le s_i^k, \quad i = k+2,\dots,n, \quad (C8: i \ne k+1) \qquad \tilde{r}_{k+1} \le s_k^k, \qquad (C8: i = k+1)$$
$$\tilde{r}_i \le s_i^1 - d^k, \quad i = 3,\dots,k, \qquad (C9: i \ne 2) \qquad \tilde{r}_2 \le s_1^1 - d^k, \qquad (C9: i = 2)$$

$$\tilde{r}_n \ge x_{n+1}, \qquad (C10: i = n)$$
$$\tilde{r}_i \ge \tilde{r}_{i+1}, \quad i = 3,\dots,k,k+2\dots,n-1, \qquad (C10: i \ne 2,k+1,n)$$
$$\tilde{r}_{j+1} - \tilde{r}_{j+2} + s_{j+1}^j - s_j^j \ge 0, \quad j \in \{1,k\}, \qquad (C10: i = 2,k+1)$$

which we denote by $(\mathbf{s}^k, \mathbf{s}^1, \tilde{\mathbf{r}}) \in \mathbf{O}(x_{n+1}, \mathbf{x}, d^k, d^1)$, where $\mathbf{s}^k = (s_n^k, \dots s_k^k)$, $\mathbf{s}^1 = (s_n^1, \dots, s_1^1)$, and $\tilde{\mathbf{r}} = (\tilde{r}_n, \dots, \tilde{r}_2)$. The optimality conditions become (1) if $s_k^k < s_{k+1}^k$, then $s_i^k = x_{n+1}$ for $i = k+1,\dots,n$ and $\tilde{r}_i = s_i^k = x_{n+1}$ for $i = k+2,\dots,n$ and $\tilde{r}_{k+1} = s_k^k$, and (2) if $s_1^1 < s_2^1$, then $s_i^1 = s_k^1$ for $i = 2,\dots,k$, $s_k^k = s_k^1 - d^k$, $\tilde{r}_i = \tilde{r}_{k+1} = s_k^k$ for $i = 3,k$, and $\tilde{r}_2 - \tilde{r}_3 + s_2^1 - s_1^1 = 0$, which implies that $\tilde{r}_2 = s_1^1 - d^k$ and $\tilde{r}_3 = s_2^2 - d^k$. Moreover, the objective function becomes

$$\hat{W}_t(x_{n+L},\dots,x_1,y,d^k,d^1,\mathbf{s}^k,\mathbf{s}^1,\tilde{\mathbf{r}}) = \delta(s_2^1 - d^k - \tilde{r}_2 + x_1 - d^1 - s_1^1) + h(\tilde{r}_2 - x_{n+1} + s_{k+1}^k - s_k^k) + \sum_{j\in\{1,k\}} l_j(s_{j+1}^j - s_j) + \beta v_{t+1}(x_{n+L} - y, \dots, x_{n+1} - x_{n+2}, \tilde{r}_n - x_{n+1}, \dots, \tilde{r}_2 - \tilde{r}_3).$$

Then, we have

$$\tilde{w}_t(x_{n+L},\dots,x_1,y,d^k,d^1) = \min_{(\mathbf{s}^k,\mathbf{s}^1,\tilde{\mathbf{r}})\in\mathbf{O}(x_{n+1},\mathbf{x},d^k,d^1)} \hat{W}_t(x_{n+L},\dots,x_1,y,d^k,d^1,\mathbf{s}^k,\mathbf{s}^1,\tilde{\mathbf{r}}).$$

Since v_{t+1} is multimodular, $v_{t+1}(x_{n+L} - y, \dots, x_{n+1} - x_{n+2}, \tilde{r}_n - x_{n+1}, \dots, \tilde{r}_2 - \tilde{r}_3)$ is submodular in $(x_{n+L},\dots,x_{n+1},\tilde{\mathbf{r}},y)$ and $\hat{W}_t(x_{n+L},\dots,x_1,y,d^k,d^1,\mathbf{s}^k,\mathbf{s}^1,\tilde{\mathbf{r}})$ is submodular in $(x_{n+L},\dots,x_1,y,\mathbf{s}^k,\mathbf{s}^1,\tilde{\mathbf{r}})$. To show $\tilde{w}_t(x_{n+L},\dots,x_1,y,d^k,d^1)$ is submodular in $(x_{n+L},\dots,x_1,y)$, it is equivalent to show that for small enough $\epsilon > 0$ and $\delta > 0$, for $i > j$,

$$\tilde{w}_t(x_{n+L},\dots,x_{i+1},x_i+\epsilon,x_{i-1},\dots,x_1,y,d^k,d^1) + \tilde{w}_t(x_{n+L},\dots,x_{j+1},x_j+\delta,x_{j-1},\dots,x_1,y,d^k,d^1)$$
$$\ge \tilde{w}_t(x_{n+L},\dots,x_{i+1},x_i+\epsilon,x_{i-1},\dots,x_{j+1},x_j+\delta,x_{j-1},\dots,x_1,y,d^k,d^1) + \tilde{w}_t(x_{n+L},\dots,x_1,y,d^k,d^1),$$

and

$$\tilde{w}_t(x_{n+L},\dots,x_{i+1},x_i+\epsilon,x_{i-1},\dots,x_1,y,d^k,d^1) + \tilde{w}_t(x_{n+L},\dots,x_1,y+\delta,d^k,d^1)$$
$$\ge \tilde{w}_t(x_{n+L},\dots,x_{i+1},x_i+\epsilon,x_{i-1},x_1,y+\delta,d^k,d^1) + \tilde{w}_t(x_{n+L},\dots,x_1,y,d^k,d^1).$$

For brevity, here we only prove the first one for $i, j \ne n+1$. The proofs of other cases are similar.

Let $(\check{\mathbf{s}}^k, \check{\mathbf{s}}^1, \check{\mathbf{r}}) = \operatorname{argmin}_{(\mathbf{s}^k,\mathbf{s}^1,\tilde{\mathbf{r}})\in\mathbf{O}(x_{n+1},\mathbf{x},d^k,d^1)} \hat{W}_t(x_{n+L},\dots,x_1,y,d^k,d^1,\mathbf{s}^k,\mathbf{s}^1,\tilde{\mathbf{r}})$. Note that $\check{s}_j^j = \check{s}_{j+1}^j$ if and only if at state $(x_{n+L} - y, x_{n+L-1} - x_{n+L}, \dots, x_1 - x_2, d^k, d^1)$, it is optimal to fulfill all the demand of Class j, and $\check{s}_j^j < \check{s}_{j+1}^j$ if and only if there are backorders of Class j demand after demand fulfillment, for $j = 1,k$. Therefore, we consider four cases: (a) $\check{s}_k^k = \check{s}_{k+1}^k$ and $\check{s}_1^1 = \check{s}_2^1$, (b) $\check{s}_k^k < \check{s}_{k+1}^k$ and $\check{s}_1^1 = \check{s}_2^1$, (c) $\check{s}_k^k = \check{s}_{k+1}^k$ and $\check{s}_1^1 < \check{s}_2^1$, and (d) $\check{s}_k^k < \check{s}_{k+1}^k$ and $\check{s}_1^1 < \check{s}_2^1$.

(a) In this case, at state $(x_{n+L} - y, x_{n+L-1} - x_{n+L}, \dots, x_1 - x_2, d^k, d^1)$, it is optimal to fulfill all the demand of Class k and Class 1. If we increase x_i by ϵ, we increase the inventory level of products with lifetime i by ϵ and reduce the inventory level of products with lifetime $i-1$ by ϵ. By the structure of the optimal allocation and disposal policies, it is still optimal to fulfill all the demand of Class k and Class 1 if we increase x_i by ϵ. In other words, if we let $(\bar{\mathbf{s}}^k, \bar{\mathbf{s}}^1, \bar{\mathbf{r}}) = \operatorname{argmin}_{(\mathbf{s}^k,\mathbf{s}^1,\tilde{\mathbf{r}})\in\mathbf{O}(x_{n+1},\mathbf{x},d^k,d^1)} \hat{W}_t(x_{n+L},\dots,x_i+\epsilon,\dots,x_1,y,d^k,d^1,\mathbf{s}^k,\mathbf{s}^1,\tilde{\mathbf{r}})$, then $\bar{s}_m^m = \bar{s}_{m+1}^m$ for $m = 1,k$. By $(C10: i = 2,k+1)$, we have $\bar{r}_{m+1} - \bar{r}_{m+2} \ge 0$ for $m = 1,k$. Similarly, if we let $(\hat{\mathbf{s}}^k, \hat{\mathbf{s}}^1, \hat{\mathbf{r}}) = \operatorname{argmin}_{(\mathbf{s}^k,\mathbf{s}^1,\tilde{\mathbf{r}})\in\mathbf{O}(x_{n+1},\mathbf{x},d^k,d^1)} \hat{W}_t(x_{n+L},\dots,x_j+\delta,\dots,x_1,y,d^k,d^1,\mathbf{s}^k,\mathbf{s}^1,\tilde{\mathbf{r}})$, then $\hat{s}_m^m = \hat{s}_{m+1}^m$ and $\hat{r}_{m+1} -$

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$\hat{r}_{m+2} \geq 0$ for $m = 1, k$. Therefore, $\bar{s}^m_{m+1} \vee \hat{s}^m_{m+1} = \bar{s}^m_m \vee \hat{s}^m_m$, $\bar{s}^m_{m+1} \wedge \hat{s}^m_{m+1} = \bar{s}^m_m \wedge \hat{s}^m_m$,

$$\bar{r}_{m+1} \vee \hat{r}_{m+1} - \bar{r}_{m+2} \vee \hat{r}_{m+2} + \bar{s}^m_{m+1} \vee \hat{s}^m_{m+1} - \bar{s}^m_m \vee \hat{s}^m_m = \bar{r}_{m+1} \vee \hat{r}_{m+1} - \bar{r}_{m+2} \vee \hat{r}_{m+2} \geq 0,$$

and

$$\bar{r}_{m+1} \wedge \hat{r}_{m+1} - \bar{r}_{m+2} \wedge \hat{r}_{m+2} + \bar{s}^m_{m+1} \wedge \hat{s}^m_{m+1} - \bar{s}^m_m \wedge \hat{s}^m_m = \bar{r}_{m+1} \wedge \hat{r}_{m+1} - \bar{r}_{m+2} \wedge \hat{r}_{m+2} \geq 0.$$

This suggests that $(\bar{\mathbf{s}}^k \vee \bar{\mathbf{s}}^k, \bar{\mathbf{s}}^1 \vee \bar{\mathbf{s}}^1, \bar{\mathbf{r}} \vee \bar{\mathbf{r}})$ and $(\bar{\mathbf{s}}^k \wedge \bar{\mathbf{s}}^k, \bar{\mathbf{s}}^1 \wedge \bar{\mathbf{s}}^1, \bar{\mathbf{r}} \wedge \bar{\mathbf{r}})$ satisfy $(C10 : i = 2, k+1)$. Notice that the new constraint set is a lattice except for $(C10 : i = 2, k+1)$. Thus, $(\bar{\mathbf{s}}^k \vee \bar{\mathbf{s}}^k, \bar{\mathbf{s}}^1 \vee \bar{\mathbf{s}}^1, \bar{\mathbf{r}} \vee \bar{\mathbf{r}})$ is a feasible solution to the problem $\min_{(\mathbf{s}^k, \mathbf{s}^1, \tilde{\mathbf{r}}) \in \mathbf{O}(x_{n+1}, \mathbf{x}, d^k, d^1)} \hat{W}_t(x_{n+L}, \ldots, x_i + \epsilon, \ldots, x_j + \delta, \ldots, x_1, y, d^k, d^1, \mathbf{s}^k, \mathbf{s}^1, \tilde{\mathbf{r}})$ and $(\bar{\mathbf{s}}^k \wedge \bar{\mathbf{s}}^k, \bar{\mathbf{s}}^1 \wedge \bar{\mathbf{s}}^1, \bar{\mathbf{r}} \wedge \bar{\mathbf{r}})$ is a feasible solution to the problem $\min_{(\mathbf{s}^k, \mathbf{s}^1, \tilde{\mathbf{r}}) \in \mathbf{O}(x_{n+1}, \mathbf{x}, d^k, d^1)} \hat{W}_t(x_{n+L}, \ldots, x_1, y, d^k, d^1, \mathbf{s}^k, \mathbf{s}^1, \tilde{\mathbf{r}})$. Therefore,

$$\begin{aligned}
&\tilde{w}_t(x_{n+L}, \ldots, x_i + \epsilon, \ldots, x_j + \delta, \ldots, x_1, y, d^k, d^1) + \tilde{w}_t(x_{n+L}, \ldots, x_1, y, d^k, d^1) \\
&\quad \leq \hat{W}_t(x_{n+L}, \ldots, x_i + \epsilon, \ldots, x_j + \delta, \ldots, x_1, y, d^k, d^1, \bar{\mathbf{s}}^k \vee \hat{\mathbf{s}}^k, \bar{\mathbf{s}}^1 \vee \hat{\mathbf{s}}^1, \bar{\mathbf{r}} \vee \hat{\mathbf{r}}) \\
&\qquad\qquad + \hat{W}_t(x_{n+L}, \ldots, x_1, y, d^k, d^1, \bar{\mathbf{s}}^k \wedge \hat{\mathbf{s}}^k, \bar{\mathbf{s}}^1 \wedge \hat{\mathbf{s}}^1, \bar{\mathbf{r}} \wedge \hat{\mathbf{r}}) \\
&\quad \leq \hat{W}_t(x_{n+L}, \ldots, x_i + \epsilon, \ldots, x_1, y, d^k, d^1, \bar{\mathbf{s}}^k, \bar{\mathbf{s}}^1, \bar{\mathbf{r}}) + \hat{W}_t(x_{n+L}, \ldots, x_j + \delta, \ldots, x_1, y, d^k, d^1, \hat{\mathbf{s}}^k, \hat{\mathbf{s}}^1, \hat{\mathbf{r}}) \\
&\quad = \tilde{w}_t(x_{n+L}, \ldots, x_i + \epsilon, \ldots, x_1, y, d^k, d^1) + \tilde{w}_t(x_{n+L}, \ldots, x_j + \delta, \ldots, x_1, y, d^k, d^1),
\end{aligned}$$

where the second inequality to due to the fact that $\hat{W}_t$ is submodular.

The proofs of cases (b), (c), and (d) are similar. This completes the proof of property (1).

(2) The proofs of (2) and (3) are similar to the proof of part (2) of Lemma 4. □

Lemma 7. *If v_{t+1} satisfies properties in Proposition 1, then the optimal ordering policy in period t is described in Theorem 3 and v_t also satisfies properties in Proposition 1.*

Proof. The optimal ordering policy follows from property (ii) of Theorem 1 in Li and Yu (2014) since $V_t(y, x_{n+L-1}, \ldots, x_1) = Ew_t(y, x_{n+L-1}, \ldots, x_1, D^k_t, D^1_t)$ is multimodular in $(y, x_{n+L-1}, \ldots, x_1)$. Property (1) of Proposition 1 follows from property (i) of Theorem 1 of Li and Yu (2014). Properties (2) and (3) follow from Envelope Theorem and properties (2) and (3) of Lemma 6. □

This completes the induction. Next we provide the proofs of the other propositions.

Proof of Proposition 2: (1) This is obviously true if $r^*_{i,t} = 0$. If $r^*_{i,t} > 0$, by its definition, we have $\frac{\partial U_t}{\partial r_i}(x_{n+L}, \ldots, x_{n+1}, z_n, \ldots, z_{i+1}, r^*_{i,t}, \mathbf{0}) = 0$ and $\frac{\partial U_t}{\partial r_{i+1}}(x_{n+L}, \ldots, x_{n+1}, z_n, \ldots, z_{i+2}, r^*_{i+1,t}, \mathbf{0}) \geq 0$. Note that $U_t(x_{n+L}, \ldots, x_{n+1}, \mathbf{r}) = h \sum_{i=2}^{n} r_i - \delta \sum_{i=2}^{n} r_i + \beta v_{t+1}(x_{n+L}, x_{n+L}, \ldots, x_{n+1}, \mathbf{r})$ is multimodular in $(x_{n+L}, \ldots, x_{n+1}, \mathbf{r})$. This implies that for any j, l, and $\epsilon \geq 0$,

$$\frac{\partial U_t}{\partial r_j}(x_{n+L}, \ldots, x_{n+1}, \mathbf{r} + \epsilon \mathbf{e}_j) \geq \frac{\partial U_t}{\partial r_j}(x_{n+L}, \ldots, x_{n+1}, \mathbf{r} + \epsilon \mathbf{e}_l), \tag{19}$$

where $\mathbf{e}_j$ ($\mathbf{e}_l$) is an $(n-1)$-dimensional vector with the jth (l^{th}) element being 1 and other elements being 0. Therefore,

$$\begin{aligned}
&\frac{\partial U_t}{\partial r_{i+1}}(x_{n+L}, \ldots, x_{n+1}, z_n, \ldots, z_{i+1} + r^*_{i,t}, \mathbf{0}) \leq \frac{\partial U_t}{\partial r_i}(x_{n+L}, \ldots, x_{n+1}, z_n, \ldots, z_{i+1} + r^*_{i,t}, \mathbf{0}) \\
&\quad \leq \frac{\partial U_t}{\partial r_i}(x_{n+L}, \ldots, x_{n+1}, z_n, \ldots, z_{i+1}, r^*_{i,t}, \mathbf{0}) = 0 \leq \frac{\partial U_t}{\partial r_{i+1}}(x_{n+L}, \ldots, x_{n+1}, , z_n, \ldots, z_{i+2}, r^*_{i+1,t}, \mathbf{0}),
\end{aligned}$$

where the first inequality is by (13) and the second inequality is by (19). By the convexity of U_t and the definition of $r^*_{i+1,t}$, we have $z_{i+1} + r^*_{i,t} \leq r^*_{i+1,t}$, or equivalently, $r^*_{i,t} \leq r^*_{i+1,t} - z_{i+1}$.

(2) Note that $U_t(x_{n+L}, \ldots, x_{n+1}, \mathbf{r})$ is multimodular in $(x_{n+L}, \ldots, x_{n+1}, \mathbf{r})$. The result then follows from the definition of $r^*_{i,t}$ and property (ii) of Theorem 1 in Li and Yu (2014).

Proof of Proposition 3: The proof is similar to the proof of Proposition 2.

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